

Tetrahedral sums of Hermitian matrices

Thomas C. Fraser¹, Matthias Christandl¹, and Anton Alekseev²

¹Department of Mathematical Sciences, University of Copenhagen, Universitetsparken 5, 2100 Denmark

²Department of Mathematics, University of Geneva, 7-9 rue du Conseil Général, 1211 Geneva 4, Switzerland

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Abstract

TODO: write an abstract

1 Introduction

This is the main document for this project. Previous drafts are located in the folder titled “old” and will not necessarily compile properly.

2 Tetrahedra & Hermitian matrices

2.1 Definition

`prob:tetrahedral_problem`

Problem 1 (Tetrahedral eigenvalues). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^d)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $d \times d$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad D + C = E, \quad A + F = E, \quad (1)$$

`{eq:tetrahedral_linear_constraints}`

and (ii) the six eigenvalue constraints

$$\lambda(A) = \alpha, \quad \lambda(B) = \beta, \quad \lambda(C) = \gamma, \quad \lambda(D) = \delta, \quad \lambda(E) = \epsilon, \quad \lambda(F) = \phi. \quad (2)$$

`{eq:tetrahedral_eigenvalue_constraints}`

2.2 Symmetries

Lemma 1 (Shifting). Let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be a given sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be given scalars and define shifted candidate eigenvalues by the formula

$$\begin{aligned} \alpha &:= \tilde{\alpha} + x, & \beta &:= \tilde{\beta} + y, & \gamma &:= \tilde{\gamma} + x + y, \\ \delta &:= \tilde{\delta} + z, & \epsilon &:= \tilde{\epsilon} + x + y + z, & \phi &:= \tilde{\phi} + y + z. \end{aligned} \quad (3)$$

`{eq:shifted_tetrahedral_eigenvalues}`

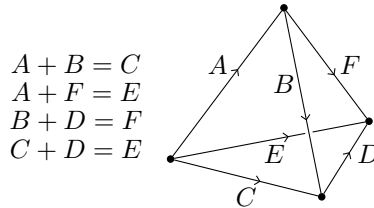


Figure 1: The six Hermitian operators appearing in the tetrahedral eigenvalue constraint problem (Problem 1), and denoted by (A, B, C, D, E, F) above, are subject to four linear constraints corresponding to the four faces of the tetrahedron with coefficients determined by the edge orientations depicted here.

`fig:tetrahedron_labelling_scheme`

Then the shifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if the unshifted eigenvalues are tetrahedral. Moreover if the unshifted eigenvalues are tetrahedral and (x, y, z) are sufficiently large, i.e.

$$x \geq \min(\tilde{\alpha}), \quad y \geq \min(\tilde{\beta}), \quad z \geq \min(\tilde{\delta}), \quad (4)$$

then the shifted spectra are both tetrahedral and non-negative.

2.3 The $d = 2$ case

In the context of the $d = 2$ tetrahedral eigenvalue problem, each traceless 2×2 matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis:

$$X = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (5)$$

The eigenvalues of X are then determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\lambda(X) = \lambda \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (6)$$

Now, for the special case of the tetrahedral eigenvalue constraint problem, one can assume the sextuple of eigenvalues are determined by a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (7)$$

Using this parameterization, it can be shown that the constraints imposed in Problem 1 precisely describe the space of all possible edge-lengths (a, b, c, d, e, f) of a genuine Euclidean tetrahedron in $\mathbb{R}^3$ [Ric02]; namely, the conjunction of a quadruplet of triangle-inequalities,

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (8)$$

together with a single non-linear constraint of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & e^2 & c^2 \\ 1 & a^2 & 0 & f^2 & b^2 \\ 1 & e^2 & f^2 & 0 & d^2 \\ 1 & c^2 & b^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (9)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when positive, is equal to square of the volume of the corresponding tetrahedron [SJS18]. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (8) and Eq. (9).

Remark 1. For 2×2 Hermitian matrices it is possible to visualize the set of eigenvalues (in the same (a, b, c, d, e, f) coordinates used above) which are known to be non-tetrahedral on the basis that they violate Eq. (34). Specifically, consider the *minimal* shifting of the traceless eigenvalues in Eq. (7), i.e.,

$$\begin{aligned} \alpha &= (2a, 0), & \beta &= (2b, 0), & \gamma &= (a + b + c, a + b - c), \\ \delta &= (2d, 0), & \epsilon &= (a + b + d + e, a + b + d - e), & \phi &= (b + d + f, b + d - f). \end{aligned} \quad (10)$$

With respect to these coordinates, Eq. (34) becomes

$$\begin{aligned} F(a, b, c, d, e, f) &:= -(a + b + c) \log(a + b + c) - (a + b - c) \log(a + b - c) \\ &\quad - (b + d + f) \log(b + d + f) - (b + d - f) \log(b + d - f) \\ &\quad + (a + b + d + e) \log(a + b + d + e) + (a + b + d - e) \log(a + b + d - e) \\ &\quad + (2b) \log(2b) \geq 0 \end{aligned} \quad (11)$$

See Fig. 3 for a depiction of the region violating Eq. (11).

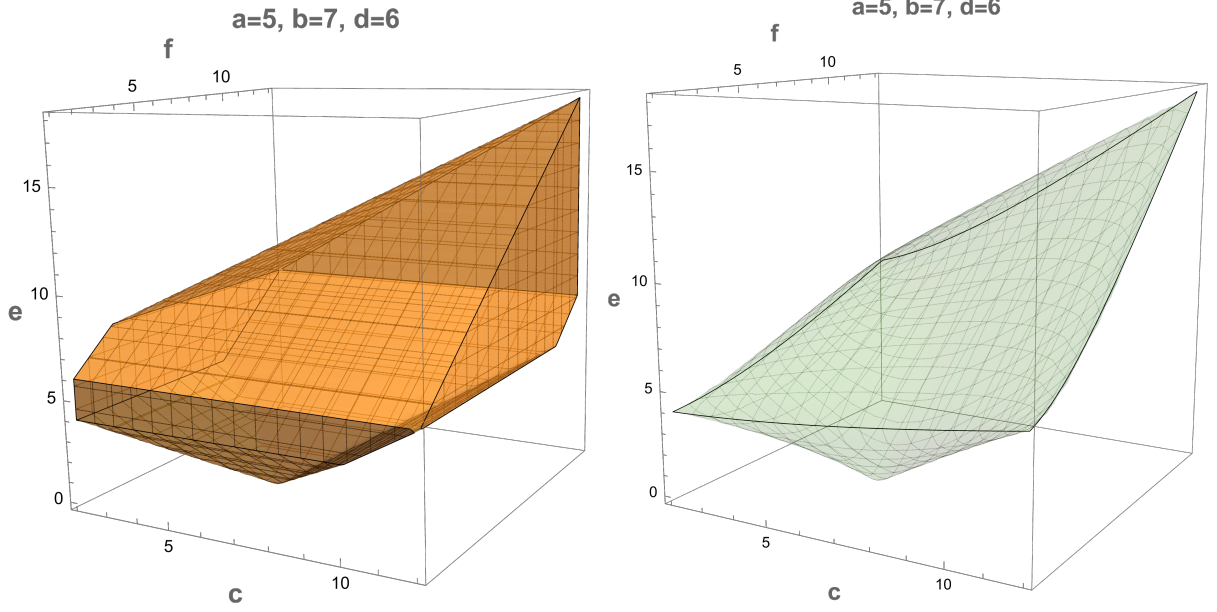


Figure 2: Projections of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (8). (Right) The set of all values (c, e, f) satisfying both Eq. (8) and Eq. (9). [fig:tetrahedron_moduli_space_example](#)

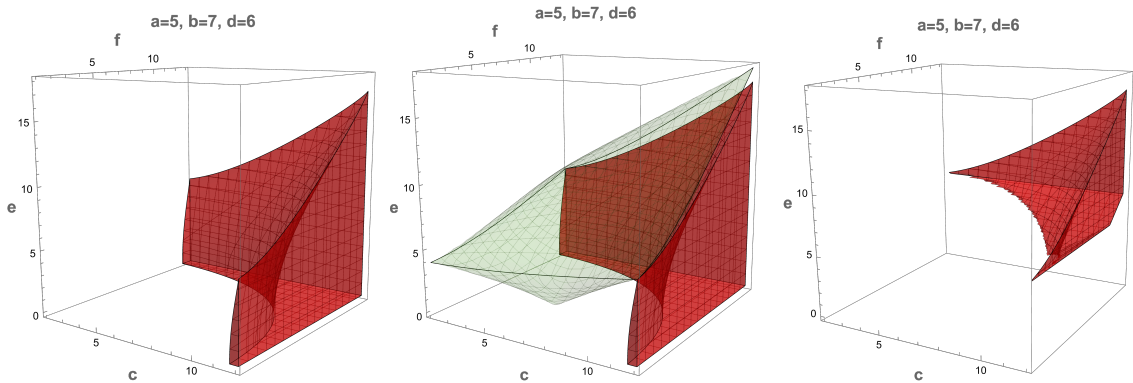


Figure 3: (Left) The region for which (c, e, f) violates Eq. (11) with $(a, b, c) = (5, 7, 6)$. (Middle) Comparing the region of tetrahedral (c, e, f) with those which violate Eq. (11). (Right) The values of (c, e, f) satisfying Eq. (8) but violating Eq. (11). [fig:ssa_extra_mileage](#)

2.4 Comparison with the CSW inequality

At this stage, it is worth comparing the inequality in Eq. (34) to a similar inequality derived by Christandl, Şahinoğlu, and Walter [CSW18], which holds for tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are non-negative and sufficiently small in the sense that $\lambda_1 + |\lambda| \leq 1$:

$$h(|\gamma|) + |\gamma| S(\bar{\gamma}) + h(|\phi|) + |\phi| S(\bar{\phi}) - h(|\beta|) - |\beta| S(\bar{\beta}) - h(|\epsilon|) - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad (\text{eq:SSA_like_CSW}) \quad (12)$$

The inequality appearing in Eq. (34) is equivalent to the following by Eq. (200):

$$-|\gamma| \log |\gamma| + |\gamma| S(\bar{\gamma}) - |\phi| \log |\phi| + |\phi| S(\bar{\phi}) + |\beta| \log |\beta| - |\beta| S(\bar{\beta}) + |\epsilon| \log |\epsilon| - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad (\text{eq:SSA_like_ineq_equiv}) \quad (13)$$

The difference between Eq. (12) and Eq. (13) is captured by a formal substitution replacing the binary entropy $h(|\lambda|)$ with the expression $-|\lambda| \log |\lambda|$. We note that it is possible to derive Eq. (13) from Eq. (12) by a downscaling argument. Specifically, while the set of tetrahedral eigenvalues is closed under scaling all eigenvalues by a non-negative r ,

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \mapsto (r\alpha, r\beta, r\gamma, r\delta, r\epsilon, r\phi), \quad (\text{eq:tetrahedral_scaling}) \quad (14)$$

the form and positivity of Eq. (12) is *not* invariant under scaling by r . Define q_r by the formula

$$q_r := r^{-1} (h(r|\gamma|) + h(r|\phi|) - h(r|\beta|) - h(r|\epsilon|)). \quad (15)$$

Letting $r \in (0, 1)$ be positive and small, we see that

$$r^{-1} h(r|\lambda|) = |\lambda| (1 - \log r) - |\lambda| \log |\lambda| + O(r^2) \quad (16)$$

which implies the following limit holds when $|\gamma| + |\phi| = |\beta| + |\epsilon|$:

$$q_0 := \lim_{r \rightarrow 0} q_r = -|\gamma| \log |\gamma| - |\phi| \log |\phi| + |\beta| \log |\beta| + |\epsilon| \log |\epsilon|. \quad (17)$$

In summary, by scaling down the eigenvalues in Eq. (12) by $r \in (0, 1)$ according to Eq. (14) and then scaling up Eq. (12) by multiplication by r^{-1} and taking the limit as $r \rightarrow \infty$, one recovers Eq. (13). Moreover, it can be shown that $q_r \geq q_0$ for all r and thus Eq. (13) is strictly stronger than Eq. (12). To prove this, we see that the trace equalities imply

$$\begin{aligned} q_r - q_0 &= r^{-1} (1 - r|\beta|) \log(1 - r|\beta|) + r^{-1} (1 - r|\epsilon|) \log(1 - r|\epsilon|) \\ &\quad - r^{-1} (1 - r|\gamma|) \log(1 - r|\gamma|) - r^{-1} (1 - r|\phi|) \log(1 - r|\phi|). \end{aligned} \quad (18)$$

Approximating $-(1-x) \log(1-x)$ by a Taylor series for $x \approx 0$,

$$-(1-x) \log(1-x) = x - \frac{x^2}{2} + O(x^3), \quad (19)$$

produces

$$q_r - q_0 = \frac{r}{2} (|\beta|^2 + |\epsilon|^2 - |\gamma|^2 - |\phi|^2) + O(r^2) = r |\alpha| |\delta| + O(r^2) \quad (20)$$

which implies $q_r > q_0$ and least for sufficiently small r . To prove $q_r > q_0$ holds for all r , by continuity, it therefore suffices to show that the gradient of $q_r - q_0$ with respect to r never changes sign for $r > 0$.

$$\begin{aligned} \partial_r (q_r - q_0) &= \frac{1}{r^2} (\log(1 - r|\gamma|) + \log(1 - r|\phi|) - \log(1 - r|\beta|) - \log(1 - r|\epsilon|)) \\ &= \frac{1}{r^2} \log \left(\frac{(1 - r|\gamma|)(1 - r|\phi|)}{(1 - r|\beta|)(1 - r|\epsilon|)} \right). \end{aligned} \quad (21)$$

Thus the gradient $\partial_r (q_r - q_0)$ vanishes only if the argument of the above logarithm is one, or equivalently when

$$0 = (1 - r|\gamma|)(1 - r|\phi|) - (1 - r|\beta|)(1 - r|\epsilon|) = r^2 (|\gamma| |\phi| - |\beta| |\epsilon|) = r^2 |\alpha| |\delta|. \quad (22)$$

As we are implicitly assuming $|\alpha| \neq 0$ and $|\delta| \neq 0$, we see that the gradient of $q_r - q_0$ never changes when $r > 0$ and thus $q_r - q_0$ is increasing with increasing $r > 0$.

3 Submodular and supermodular inequalities on positive semidefinite matrices

[lem:quadratic_ineq](#)

Lemma 2. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then*

$$\sum_j \lambda_j(A+B)^2 + \sum_j \lambda_j(B+C)^2 \leq \sum_j \lambda_j(B)^2 + \sum_j \lambda_j(A+B+C)^2. \quad (23)$$

Proof. Note that $\sum_j \lambda_j(X)^2 = \text{Tr}[X^2]$ for any Hermitian operator X . Therefore,

$$\begin{aligned} & \text{Tr}[(B)^2] + \text{Tr}[(A+B+C)^2] - \text{Tr}[(A+B)^2] - \text{Tr}[(B+C)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[AB] + 2\text{Tr}[BC] + 2\text{Tr}[AC]) \\ &\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2\text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[BC]) \\ &= 2\text{Tr}[AC]. \end{aligned} \quad (24)$$

Since $A, C \geq 0$ are both positive semidefinite, we also have $\text{Tr}[AC] \geq 0$ which proves the claim. $\square$

Bouhrou, Gaubert, and Sagnol claim to prove the following family of inequalities [\[BGS08, Lemma 5.3\]](#). [lem:bouhrou_ineq](#)

Lemma 3. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let $p \in [0, 1]$. Then*

$$\sum_j \lambda_j(A+B)^p + \sum_j \lambda_j(B+C)^p \geq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A+B+C)^p. \quad (25)$$

The following operator inequality can be used to prove a number of inequalities [\[TCL11, Lemma 2.2\]](#). [lem:mme_operator_ineq](#)

Lemma 4. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A+B+C)^{\otimes k} + B^{\otimes k} \geq (A+B)^{\otimes k} + (B+C)^{\otimes k}. \quad (26)$$

Proof. The result is obtained from applying a non-commutative multinomial expansion and noting that every summand on the right-hand-side above also appears on the left-hand-side above. $\square$

To obtain an inequality which depends only on the eigenvalues of $B, A+B, B+C$, and $A+B+C$, one can simply pair the positive semidefinite operator in Lemma 4 with an $\text{SU}(d)$ -equivariant and positive semidefinite operator:

[cor:isotypic_ineq](#)

Corollary 5. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let $\lambda \vdash k$ be a Young diagram with k boxes. Then*

$$\text{Tr}[P_\lambda(A+B+C)^{\otimes k}] + \text{Tr}[P_\lambda B^{\otimes k}] \geq \text{Tr}[P_\lambda(A+B)^{\otimes k}] + \text{Tr}[P_\lambda(B+C)^{\otimes k}]. \quad (27)$$

Paksoy, Turkmen, and Zhang proved a family of polynomial inequalities which are generally not functions of just the eigenvalues [\[PTZ14\]](#). These inequalities are in terms a so-called generalized matrix function first defined by Marcus and Minc [\[MM65\]](#).

Definition 1 (Generalized matrix function). Let H be a subgroup of S_n and let χ be an irreducible character of H . The *generalized matrix function* for the character χ of H is

$$d_\chi^H(X) = \sum_{\sigma \in H} \chi(\sigma) \prod_i X_{i, \sigma(i)} \quad (28)$$

The notion of a *generalized matrix function* generalizes the notion of an *immanant* (when $H = S_n$) which, in turn, contains as special cases the notion of a permanent and determinant [\[HM21\]](#). [thm:ptz_ineqs](#)

Theorem 6. *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices and let H be a subgroup of S_n with irreducible character χ . Then*

$$d_\chi^H(A+B+C) + d_\chi^H(B) \geq d_\chi^H(A+B) + d_\chi^H(B+C). \quad (29)$$

Proof. See [PTZ14]. □

As a corollary of Theorem 6, we obtain the following determinant and permanent inequalities.

Corollary 7. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Then*

$$\det(A + B + C) + \det(B) \geq \det(A + B) + \det(B + C), \quad (30)$$

$$\text{per}(A + B + C) + \text{per}(B) \geq \text{per}(A + B) + \text{per}(B + C). \quad (31)$$

Note that with the exception of the above determinant inequality, the inequalities provided by Theorem 6 are generally not equivalent to inequalities involving only eigenvalues.

One also has another determinant inequality which is non-linear in determinants [TCL11, Lemma 2.3]. *lem:det_prod_ineq*

Lemma 8. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Then*

$$\det(A + B + C) \det(B) \leq \det(A + B) \det(B + C). \quad (32)$$

Note that Lemma 8 can be seen as a special of Lemma 3 in the limit as $p \rightarrow 0$; the derivative of Lemma 3 in p at $p = 0$ yields

$$\sum_j \log \lambda_j(A + B) + \sum_j \log \lambda_j(B + C) \geq \sum_j \log \lambda_j(B) + \sum_j \log \lambda_j(A + B + C), \quad (33)$$
{eq:bouhtou_ineq_p_zero}

which is equivalent to Lemma 8. *thm:ssa_ineq*

Theorem 9. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then*

$$\sum_j \lambda_j(A + B) \log \lambda_j(A + B) + \sum_j \lambda_j(B + C) \log \lambda_j(B + C) \quad (34)$$
{eq:ssa_like_ineq}

$$\leq \sum_j \lambda_j(B) \log \lambda_j(B) + \sum_j \lambda_j(A + B + C) \log \lambda_j(A + B + C). \quad (35)$$

The most comprehensive result concerning the eigenvalues of partial sums of Hermitian matrices we have encountered thus far is [Sag13, Proposition 2.3] which provides the following result. *thm:sagnol_ineq*

Theorem 10. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ such that the derivative of f is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds*

$$\text{Tr } f(A + B + C) + \text{Tr } f(B) \geq \text{Tr } f(A + B) + \text{Tr } f(B + C). \quad (36)$$
{eq:sagnol_ineq}

Proof. The following proof is a rough summary of the proof found in [Sag13, Section 2.1]. Let X, Y be positive semidefinite and consider following difference:

$$F(Y|X) := \text{Tr } f(X + Y) - \text{Tr } f(Y). \quad (37)$$

The inequality in Eq. (36) is thus equivalent to $F(C|B) \leq F(A + C|B)$. To prove Eq. (36) it suffices to prove that the value of $F(Y|X)$ is a nondecreasing function under $Y \mapsto Y + Z$ for arbitrary positive semidefinite X, Y , and Z . To establish the claim, let $\epsilon > 0$ be small and let $Z = \epsilon W$ for non-zero positive semidefinite W . It can be shown¹ that

$$(\partial_Y F(Y|X))(W) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F(Y + \epsilon W|X) - F(Y|X)) \quad (38)$$

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{Tr } f(X + Y + \epsilon W) - \text{Tr } f(Y + \epsilon W) - \text{Tr } f(X + Y) + \text{Tr } f(Y)) \quad (39)$$

$$= \text{Tr } [(f'(X + Y) - f'(Y)) W]. \quad (40)$$

Using the assumption that f' is an operator monotone together with $X + Y \geq Y$ (because $X \geq 0$), we have $f'(X + Y) \geq f'(Y)$ and thus $f'(X + Y) - f'(Y) \geq 0$. Therefore $(\partial_Y F(Y|X))(W) \geq 0$ for all W and thus $F(Y|X)$ is nondecreasing with respect to $Y \mapsto Y + Z$ as desired. □

¹The statement follows, without loss of generality, by assuming that all the matrices involved are positive *definite*. This assumption is permitted because f is assumed continuous and the eigenvalues of a matrix are also continuous functions (see [Sag13] for a detailed explanation).

As corollaries one obtains some of the previously mentioned inequalities for various functions f satisfying the conditions of the theorem; for example, choosing $f(x) = x \log x$ yields Theorem 9, while choosing $f(x) = -x^p$ for $p \in (0, 1]$ yields Lemma 3, and choosing $f(x) = x^p$ for $p \in (1, 2]$ yields the following.

Corollary 11. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let $p \in (1, 2]$. Then*

$$\sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \leq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (41)$$

TC: On Sept. 18th, I came across an interesting discussion on [mathoverflow](https://mathoverflow.net/questions/257562/a-determinant-inequality) <https://mathoverflow.net/questions/257562/a-determinant-inequality> which appears deeply related to the topic of submodular inequalities. In that post the following result is stated without explicit proof and is based on the theory of so-called “hyperbolic polynomials” p and their real roots λ_i . There are a handful of conjectured inequalities on the page.

Remark 2. There is also the paper by Lami et al. which investigates submodular inequalities of functions of blocks of positive matrices [Lam+17].

4 The additive quantum entropy cone

4.1 Discovering the direct sum quantum entropy cone

One of the motivations for studying inequalities between von Neumann entropies of subsystems of multipartite quantum states is because the closure $\bar{\mathcal{M}}_n$ of the set of possible *allocations of von Neumann entropies*,

$$\mathcal{M}_n := \{(H_{\text{vN}}(\text{Tr}_{[n] \setminus S}(\rho_{12 \dots n})))_{\emptyset \neq S \subseteq [n]} \mid \rho_{12 \dots n} \in \mathcal{D}(\mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n)\}, \quad (42)$$

was shown by Pippenger to be a convex cone and thus it suffices to restrict ones attention to linear inequalities of von Neumann entropies [Pip03].

One might also be interested in set of possible allocations of some other function f of marginals of quantum states, or in our case, the possible allocations of some other function when evaluated over all *partial sums* of a tuple of positive semidefinite Hermitian matrices. To this end, we consider the following set

$$\mathcal{A}_{f,n} := \left\{ \left(f \left(\sum_{i \in S} X_i \right) \right)_{\emptyset \neq S \subseteq [n]} \mid X_1, \dots, X_n \geq 0 \right\}. \quad (43)$$

Question: *Does there exist a function f such that $\bar{\mathcal{A}}_{f,n}$ forms a convex cone?*

Following the argument due to Pippenger, the closure $\bar{\mathcal{A}}_{f,n}$ would form a convex cone if and only if $\mathcal{M}_{f,n}$ was both *additive* and *approximately dilutable*. Ignoring the condition of approximate dilutability altogether, what would it take to prove that $\mathcal{M}_{f,n}$ was additive? A constructive proof would be the following: given two n -tuples $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ with allocations $\mathcal{M}_{f,n}(X)$ and $\mathcal{M}_{f,n}(Y)$, compute a new n -tuple $Z = (Z_1, \dots, Z_n)$ with allocation $\mathcal{M}_{f,n}(Z)$ such that

$$\mathcal{M}_{f,n}(Z) = \mathcal{M}_{f,n}(X) + \mathcal{M}_{f,n}(Y), \quad (44)$$

or in other words,

$$\forall S \subseteq [n] \quad : \quad f \left(\sum_{i \in S} Z_i \right) = f \left(\sum_{i \in S} X_i \right) + f \left(\sum_{i \in S} Y_i \right). \quad \text{\textcolor{blue}{\{eq:additivity_partial_sums\}}} \quad (45)$$

In [Pip03], Pippenger used two properties of von Neumann entropies to prove additivity of $\mathcal{A}_n$: (i) they are additive across tensor products, i.e. $H(\rho \otimes \sigma) = H(\rho) + H(\sigma)$, and (ii) they interact well with partial tracing over subsystems. In our context, suppose a similar argument worked; let $\odot$ be some binary operation on Hermitian matrices such that f was additive with respect to this operation:

$$\forall X, Y : f(X \odot Y) = f(X) + f(Y). \quad (46)$$

There are numerous pairs $(f, \odot)$ of functions and binary operations which satisfy the above equation: e.g., entropies and tensor products $(H, \otimes)$, traces and direct sums $(\text{Tr}, \oplus)$, log traces/determinants and tensor products $(\log \text{Tr}, \otimes)$, $(\log \det, \otimes)$, etc.

A potential difficulty arises when trying to find a function/operation pair which additionally satisfies Eq. (45). If f is additive across some $\odot$ and Eq. (45) holds, then we could conclude that $\mathcal{M}_{f,n}$ was additive. Exploring this possibility further, to prove additivity of $\mathcal{M}_{f,n}$, it suffices to find an n -tuple $Z = (Z_1, \dots, Z_n)$ satisfying the equations

$$\forall S \subseteq [n] \quad : \quad \sum_{k \in S} Z_k = \left(\sum_{i \in S} X_i \right) \odot \left(\sum_{j \in S} Y_j \right). \quad \text{\textcolor{blue}{\{eq:proposed_constraint\}}} \quad (47)$$

In other words, the new n -tuple $Z = (Z_1, \dots, Z_n)$ is determined by the binary operation $\odot$ because $Z_k = X_k \odot Y_k$. Therefore, the binary operation $\odot$ would need to satisfy

$$\forall S \subseteq [n] \quad : \quad \sum_{k \in S} (X_k \odot Y_k) = \left(\sum_{i \in S} X_i \right) \odot \left(\sum_{j \in S} Y_j \right). \quad (48)$$

This above equation is the “additive analogue” of the fact that tensor products and partial traces are compatible meaning

$$\forall S \subseteq [n] \quad : \quad \text{Tr}_S (\rho_{12\dots n}^X \otimes \rho_{12\dots n}^Y) = \text{Tr}_S (\rho_{12\dots n}^X) \otimes \text{Tr}_S (\rho_{12\dots n}^Y). \quad (49)$$

To determine which binary operation $\odot$ is needed, consider an instance of Eq. (47) corresponding to subset S such that $|S| = 2$:

$$(X_1 \odot Y_1) + (X_2 \odot Y_2) = (X_1 + X_2) \odot (Y_1 + Y_2). \quad (50)$$

From here it is clear that $\odot$ can be taken to be the direct sum $\oplus$ and the function f any function of Hermitian matrices which is the trace of some function of eigenvalues, i.e. $f = \text{Tr} \circ g$. In this setting, the desired relation follows:

$$f(X \odot Y) = \text{Tr}(g(X \oplus Y)) = \text{Tr}(g(X) \oplus g(Y)) = \text{Tr}(g(X)) + \text{Tr}(g(Y)) = f(X) + f(Y). \quad (51)$$

We now return to the question of approximate dilutability.

Followup question: *Does there exist a function g such that $\mathcal{M}_{\text{Tr} \circ g, n}$ is approximately dilutable?*

In order for $\bar{\mathcal{M}}_{\text{Tr} \circ g, n}$ to form a convex cone, it must be that $\mathcal{M}_{\text{Tr} \circ g, n}$ is both additive and approximately dilutable. Fortunately, there exists choices for g such that approximate dilutability holds.

For example, let $g(x) = -x \log x$. Take $\mu > 0$ and note that²

$$\begin{aligned} H(\mu X) &= \text{Tr}(g(\mu X)) \\ &= - \sum_j \lambda_j(\mu X) \log \lambda_j(\mu X) \\ &= -\mu \sum_j \lambda_j(X) \log \lambda_j(\mu X) \\ &= -\mu \sum_j \lambda_j(X) \log \lambda_j(X) - \mu \sum_j \lambda_j(X) \log \mu \\ &= \mu \text{Tr}(g(X)) - \mu \log \mu \text{Tr}(X) \\ &= \mu H(X) - \mu \log \mu \text{Tr}(X). \end{aligned}$$

Therefore, for any $C > 0$ and $\epsilon > 0$, one can take $\delta > 0$ sufficiently small such that $-\mu \log \mu < C\epsilon$ holds for all $\mu \leq \delta$ implying

$$|\text{Tr}(g(\mu X)) - \mu \text{Tr}(g(X))| = |-\mu \log \mu \text{Tr}(X)| \leq C \text{Tr}(X) \epsilon \leq \epsilon, \quad (52)$$

where the last inequality follows from choosing $C = \text{Tr}(X)^{-1}$ for all partial sums $X = \sum_{i \in S} X_i$ under consideration. For another example, let $h_\beta(x) = \beta x - x \log x$, then

$$\text{Tr}(h_\beta(X)) = \beta \text{Tr}(X) + \text{Tr}(g(X)), \quad (53)$$

which implies for $\mu > 0$

$$\text{Tr}(h_\beta(\mu X)) = \beta \mu \text{Tr}(X) + \text{Tr}(g(\mu X)) = \beta \mu \text{Tr}(X) + \mu \text{Tr}(g(X)) - \mu \log \mu \text{Tr}(X) = \mu \text{Tr}(h_\beta(X)) - \mu \log \mu \text{Tr}(X). \quad (54)$$

In other words, shifting the function $\text{Tr} \circ g$ by any multiple of the trace also produces a set $\mathcal{M}_{\text{Tr} \circ h_\beta, n}$ which is approximately dilutable (and also additive). Of course, this is not surprising given that partial sum of matrices always satisfy certain linear constraints (namely $\text{Tr}(X_{S \sqcup T}) = \text{Tr}(X_S) + \text{Tr}(X_T)$ for all disjoint $S, T \subseteq [n]$) and thus line in a linear subspace.

Remark 3. There are two related formulations of the idea of partial sums of positive semidefinite Hermitian matrices. The first formulation is the formulation presented in this section; specifically, one is interested in Hermitian matrices X which are positive semidefinite $X \geq 0$ and small in the sense that $\|X\|_\infty \leq 1$ and more generally $\|X\|_\infty < \epsilon$ for arbitrarily small $\epsilon > 0$. The second formulation aims to instead normalize all

²The function $g(x) = -x \log x$ applied to positive semidefinite Hermitian matrices satisfies $\text{Tr}(g(\mu X)) = \mu \text{Tr}(g(X)) + g(\mu) \text{Tr}(X)$.

of the non-zero matrices involved by letting $X = \text{Tr}(X)\bar{X}$ where $\bar{X} = X/\text{Tr}(X)$. The consequences of this normalisation scheme is two-fold. First the function $f = \text{Tr} \circ g$ can be reexpressed by setting $\mu = \text{Tr}(X)^{-1}$ to obtain

$$\text{Tr}(g(\bar{X})) = \frac{\text{Tr}(g(X))}{\text{Tr}(X)} + \log \text{Tr}(X) \quad (55)$$

or equivalently

$$\text{Tr}(g(X)) = \text{Tr}(X) \text{Tr}(g(\bar{X})) - \text{Tr}(X) \log \text{Tr}(X). \quad (56)$$

The second consequence is that partial sums become

$$\sum_{i \in S} X_i = \sum_{i \in S} \text{Tr}(X_i) \bar{X}_i \quad (57)$$

which implies that *normalized* partial sums can be expressed as convex combinations:

$$\bar{X}_S = \frac{\sum_{i \in S} X_i}{\text{Tr}(\sum_{i \in S} X_i)} = \sum_{i \in S} \frac{\text{Tr}(X_i)}{\text{Tr}(\sum_{i \in S} X_i)} \bar{X}_i = \sum_{i \in S} p_i \bar{X}_i \quad (58)$$

where $p_i = \text{Tr}(X_i)/\text{Tr}(\sum_{i \in S} X_i)$. The second formulation would then define allocations of entropies of ensembles $\mathcal{E} = \{(p_i, \rho_i)\}_{i \in [n]}$ of density matrices by setting

$$\alpha_{\mathcal{E}}(S) = H(\sum_{i \in S} p_i \rho_i) = p_S H\left(\sum_{i \in S} \frac{p_i}{p_S} \rho_i\right) - p_S \log p_S \quad (59)$$

where $p_S = \sum_{i \in S} p_i$ and $H(X) = -\sum_j \lambda_j(X) \log \lambda_j(X)$ is the von Neumann entropy function (freely extended to consider subnormalized matrices).

4.2 The additive entropy cone

The purpose of this section is to introduce the notion of an “additive” quantum entropy cone which serves as a kind of analogy to the ordinary quantum entropy cone considered by Pippenger [Pip03]. Formally, quantum states ρ_{ABC} and their reduced states (ρ_{AB} , etc.) are replaced by tuples of positive semidefinite operators (X, Y, Z) and their partial sums ($X + Y$, etc.). Moreover the von Neumann entropy of each reduced state is replaced by an unnormalized entropy of each partial sum. Using essentially the same approach adopted by [Pip03], we prove that the topological closure of the set of all such “allocations” of unnormalized entropies of partial sums forms a convex cone, which we term the “additive quantum entropy cone”. Moreover, we prove inequalities constraining the boundary of the additive quantum entropy cone which parallel the basic positivity, subadditivity and strong subadditivity inequalities which constrain the boundary of the ordinary quantum entropy cone.

4.2.1 The shifted unnormalized entropy function

The primary quantity of interest will be a real-valued function of eigenvalues which resembles von Neumann entropies.

Definition 2. Let $\beta \in \mathbb{R}$ be a constant and define a continuous function $h_\beta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ by

$$h_\beta(x) = \begin{cases} \beta x - x \log x & x > 0, \\ 0 & x = 0, \end{cases} \quad (60)$$

and let H_β be the real-valued function on PSD matrices given by

$$H_\beta(X) = \text{Tr}(h_\beta(X)) = \sum_{i=1}^d h_\beta(\lambda_i(X)). \quad \text{\textcolor{blue}{\code{eq:shifted_unnormalized_entropy}}} \quad (61)$$

The quantity $H_\beta(X)$ will be called the **(shifted) unnormalized entropy** of X .

Remark 4. For the ease of notation, we will henceforth adopt the convention that $0 \log 0 := 0$ so that one can write $h_\beta(x) = \beta x - x \log x$ without consideration for the $x = 0$ case explicitly.

Remark 5. It is worth remarking why H_β is termed the (shifted) unnormalized entropy. Suppose $(\lambda_1, \dots, \lambda_d)$ are the eigenvalues of X . When $\beta = 0$, we have the following

$$H_0(X) = - \sum_{i=1}^d \lambda_i \log \lambda_i. \quad (62)$$

We elect to call $H_0(X)$ the *unnormalized* entropy of X because it coincides with the von Neumann entropy when restricted inputs where $\text{Tr}(X) = 1$ (i.e. when X is a density matrix). In general, $H_0(X)$ is related to the von Neumann entropy of the *normalization* of X according to the formula

$$H_{\text{vN}}\left(\frac{X}{\text{Tr}(X)}\right) = H_0\left(\frac{X}{\text{Tr}(X)}\right) = \frac{1}{\text{Tr}(X)} H_0(X) + \log \text{Tr}(X), \quad (63)$$

or equivalently,

$$H_0(X) = \text{Tr}(X) H_0\left(\frac{X}{\text{Tr}(X)}\right) - \text{Tr}(X) \log \text{Tr}(X). \quad (64)$$

The term $H_\beta(X)$ is called the *shifted* or β -*shifted* unnormalized entropy because $H_\beta(X) = H_0(X) + \beta \text{Tr}(X)$.

The following lemma lists some basic properties of the function H_β .

Lemma 12. Let $\beta \in \mathbb{R}$, let H_β be as defined by Eq. (61) and let X be a PSD matrix.

- H_β is a spectral function: for all unitary matrices U ,

$$H_\beta(X) = H_\beta(UXU^\dagger). \quad (65)$$

- If $(\lambda_1, \dots, \lambda_d)$ are the eigenvalues of X , then

$$H_\beta(X) = \sum_{i=1}^d (\beta \lambda_i - \lambda_i \log \lambda_i). \quad (66)$$

- Let 0_d be the $d \times d$ zero matrix. Then

$$H_\beta(0_d) = 0 \quad (67)$$

- Let I_d be the $d \times d$ identity matrix. Then

$$H_\beta(I_d) = \beta d \quad (68)$$

It is important to note that $H_\beta(X)$ is not a homogeneous function in X meaning in general

$$H_\beta(X) \neq H_\beta(\gamma). \quad (69)$$

Nevertheless, there is a sense in which it is homogeneous up to a modification to the value of β which depends on the scaling parameter γ .

lem:deformation_homogeneous

Lemma 13. Let $\gamma > 0$ be non-zero. Then H_β is deformation-homogeneous in the sense that

eq:deformation_homogeneous

$$H_\beta(\gamma X) = \beta \gamma \operatorname{Tr}(X) + \gamma H_0(X) - \gamma \log \gamma \operatorname{Tr}(X) = \gamma H_{\beta - \log \gamma}(X). \quad (70)$$

As a special case, letting $\beta = \log \gamma$ produces

$$H_0(X) = \frac{1}{\gamma} H_{\log \gamma}(\gamma X), \quad (71)$$

and as a further special case, letting $\gamma = \operatorname{Tr}(X)^{-1}$ yields

$$H_0(X) = \operatorname{Tr}(X) H_{-\log \operatorname{Tr}(X)} \left(\frac{X}{\operatorname{Tr}(X)} \right). \quad (72)$$

4.2.2 Inequalities for the unnormalized entropy function

The first inequality is the most basic. It identifies that for sufficiently small PSD matrices X , $H_\beta(X)$ is non-negative.

Lemma 14. Let X be a PSD matrix with sufficiently small maximum eigenvalue such that

eq:norm_beta_cond

$$\|X\|_\infty \leq \exp(\beta). \quad (73)$$

Then,

$$H_\beta(X) \geq 0, \quad (74)$$

with equality (for X satisfying Eq. (73)) holding only if X has eigenvalues in the set $\{0, \exp(\beta)\}$.

Proof. If $(\lambda_1, \dots, \lambda_d)$ are the eigenvalues of X , then $H_\beta(X)$ equals

eq:eigen_sum_Hbeta

$$H_\beta(X) = \sum_{i=1}^d \lambda_i (\beta - \log \lambda_i). \quad (75)$$

Under the conditions that $X \geq 0$ is PSD and satisfying Eq. (73), we have

$$\lambda_i \geq 0, \quad \text{and} \quad \beta - \log \lambda_i \geq 0, \quad (76)$$

respectively. Therefore each summand in Eq. (75) is positive and thus $H_\beta(X) \geq 0$ holds. To prove the claim regarding equality $H_\beta(X)$ we note that each summand in Eq. (75) must be zero and thus either $\lambda_i = 0$ or $\lambda_i = \exp(\beta)$. $\square$

Remark 6. TC: If is unclear at the moment whether it is possible for X to satisfy $H_\beta(X) = 0$ while having eigenvalues different from $\{0, \exp(\beta)\}$ if one allows for $\|X\|_\infty > \exp(\beta)$. thm:unnormalized_entropy_ssa

Theorem 15. Let $\beta \in \mathbb{R}$ be arbitrary and let $X, Y, Z \geq 0$ be PSD matrices. Then H_β is both subadditive {eq:subadditive_Hbeta}

$$H_\beta(X + Y) \leq H_\beta(X) + H_\beta(Y), \quad (77)$$

and also strongly subadditive in the sense that

$$H_\beta(X + Y + Z) + H_\beta(Y) \leq H_\beta(X + Y) + H_\beta(Y + Z). \quad (78)$$

Proof. We begin by noting that subadditivity (Eq. (77)) is a special case of strong subadditivity (Eq. (78)) under the substitution $(X, Y, Z) \mapsto (X, 0, Y)$ because $H_\beta(0) = 0$ for any value of β . Therefore it suffices to just prove strong subadditivity (Eq. (78)). Furthermore, it suffices to prove just the $\beta = 0$ case of Eq. (78) because $H_\beta(X) = \beta \operatorname{Tr}(X) + H_0(X)$ and {eq:balanced}

$$\operatorname{Tr}(X + Y + Z) + \operatorname{Tr}(Y) = \operatorname{Tr}(X + Y) + \operatorname{Tr}(Y + Z). \quad (79)$$

In any case, strong subadditivity of H_β or arbitrary β holds as a special case of Theorem 10 where the function f is taken to be the function

$$f(x) := -h_\beta(x) = -\beta x + x \log x. \quad (80)$$

The derivative of $-h_\beta$ is

$$(-h_\beta)'(x) = 1 - \beta + \log x \quad (81)$$

which is known to be operator monotone [Cha13] on $\mathbb{R}_{>0}$ as required. □

4.2.3 Comparison to concavity of von Neumann entropy

At this stage it is worth comparing how the inequalities in Theorem 15 compare to known inequalities concerning von Neumann entropy. Suppose ρ_X and ρ_Y are two density matrices and $\lambda \in [0, 1]$ is a convex parameter. Then it is known that the von Neumann entropy satisfies *concavity*:

$$\lambda H_{\text{vN}}(\rho_X) + (1 - \lambda) H_{\text{vN}}(\rho_Y) \leq H_{\text{vN}}(\lambda \rho_X + (1 - \lambda) \rho_Y), \quad (82)$$

but also an associated upper bound:

$$H_{\text{vN}}(\lambda \rho_X + (1 - \lambda) \rho_Y) \leq \lambda H_{\text{vN}}(\rho_X) + (1 - \lambda) H_{\text{vN}}(\rho_Y) + h_{\text{bin}}(\lambda) \quad (83)$$

where $h_{\text{bin}}(\lambda)$ is the binary entropy function:

$$h_{\text{bin}}(\lambda) = -\lambda \log \lambda - (1 - \lambda) \log \lambda. \quad (84)$$

(Upper bound): We first show that this upper bound is related to subadditivity of H_0 . If we let $p_X, p_Y \geq 0$ be weights such that

$$\lambda = \frac{p_X}{p_X + p_Y}, \quad 1 - \lambda = \frac{p_Y}{p_X + p_Y}, \quad (85)$$

then the upper bound above can be equivalently written in terms of the unnormalized entropy H_0 of

$$\sigma_X := p_X \rho_X, \quad \sigma_Y := p_Y \rho_Y, \quad (86)$$

as follows

$$H_0(\sigma_X + \sigma_Y) \leq H_0(\sigma_X) + H_0(\sigma_Y). \quad (87)$$

We note that this expression is equivalent to the subadditivity of H_0 (Eq. (77) from Theorem 15). This is because every non-zero PSD matrix σ_X can be equivalently expressed in terms of a positive weight p_X and a density matrix ρ_X according to the formula:

$$(p_X, \rho_X) := \left(\operatorname{Tr}(\sigma_X), \frac{\sigma_X}{\operatorname{Tr}(\sigma_X)} \right). \quad (88)$$

(Lower bound/concavity): It is also possible to express the concavity of von Neumann entropy as a kind of superadditivity of a different function, namely the function

$$g(X) = \text{Tr}(X) H_{\text{vN}}\left(\frac{X}{\text{Tr}(X)}\right) = H_0(X) + \text{Tr}(X) \log \text{Tr}(X). \quad (89)$$

In this notation, the concavity of von Neumann entropy is equivalent to

$$g(\sigma_X + \sigma_Y) \geq g(\sigma_X) + g(\sigma_Y). \quad (90)$$

TC: At the moment it is unclear if the function g satisfies a kind of strong superadditivity or not, but its probably worth finding out.

4.2.4 The additive unnormalized entropy cone

We are now in a position to actually define allocations of unnormalized entropies of partial sums.

Definition 3. Let $\mathbf{X} := (X_1, \dots, X_n)$ be an n -tuple of positive semidefinite Hermitian matrices. Define the **allocation of unnormalized entropies** of $\mathbf{X}$ to be the element of $\mathbb{R}^{2^n}$ given by

$$\alpha_{\beta,n}(\mathbf{X}) = \left(H_{\beta}\left(\sum_{i \in S} X_i\right) \right)_{S \subseteq [n]}. \quad (91)$$

The set of all possible allocations of unnormalized entropies will be denoted by $\mathcal{A}_{\beta,n}$.

Example 1. For example, when $n = 3$, the allocation of unnormalized entropies for $\mathbf{X} = (X, Y, Z)$ is

$$\alpha_{\beta,3}(X, Y, Z) = (0, H_{\beta}(X), H_{\beta}(Y), H_{\beta}(Z), H_{\beta}(Y + Z), H_{\beta}(X + Z), H_{\beta}(X + Y), H_{\beta}(X + Y + Z)). \quad (92)$$

Our goal will be to prove that the closure, $\bar{\mathcal{A}}_{\beta,n}$, of the set of possible allocations of unnormalized entropies is a convex cone. Following [Pip03], the closure $\bar{\mathcal{A}}_{\beta,n}$ is a convex cone if and only if $\mathcal{A}_{\beta,n}$ is both *additive* and *approximately dilutable*. To begin we show that $\mathcal{A}_{\beta,n}$ additive (closed under addition). lem: additive

Lemma 16. If $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{X}' = (X'_1, \dots, X'_n)$ are n -tuples of PSD matrices with allocations $\alpha_{\beta,n}(\mathbf{X})$ and $\alpha_{\beta,n}(\mathbf{X}')$, then the n -tuple

$$\mathbf{X} \oplus \mathbf{X}' := (X_1 \oplus X'_1, \dots, X_n \oplus X'_n), \quad (93)$$

has allocation given by

$$\alpha_{\beta,n}(\mathbf{X} \oplus \mathbf{X}') = \alpha_{\beta,n}(\mathbf{X}) + \alpha_{\beta,n}(\mathbf{X}'). \quad (94)$$

In other words, the set $\mathcal{A}_{\beta,n}$ is closed under addition.

Proof. This follows directly from the fact that H_{β} is additive across direct sums,

$$H_{\beta}(X \oplus X') = H_{\beta}(X) + H_{\beta}(X'), \quad (95)$$

and the fact that the direct sum is distributive across sums so that

$$\sum_{i \in S} X_i \oplus X'_i = \left(\sum_{j \in S} X_j \right) \oplus \left(\sum_{k \in S} X'_k \right). \quad (96)$$

□

Next we show that $\mathcal{A}_{\beta,n}$ is approximately dilutable. lem: approx_dilute

Lemma 17. The $\mathcal{A}_{\beta,n}$ is approximately dilutable.

Proof. Take $\mu > 0$ and consider the allocation of $X \geq 0$ compared to the allocation of $\mu X \geq 0$. By

$$H_\beta(\mu X) = \beta \operatorname{Tr}(\mu X) + H_0(\mu X), \quad (97)$$

$$= \beta \mu \operatorname{Tr}(X) + \mu H_0(X) - \operatorname{Tr}(X) \mu \log \mu, \quad (98)$$

$$= \mu H_\beta(X) - \mu \log \mu \operatorname{Tr}(X). \quad (99)$$

Therefore, for any constant $C > 0$ and $\epsilon > 0$, one can take $\delta > 0$ sufficiently small such that $-\mu \log \mu < C\epsilon$ holds for all $\mu \leq \delta$ implying

$$|H_\beta(\mu X) - \mu H_\beta(X)| = |-\mu \log \mu \operatorname{Tr}(X)| \leq C\epsilon \operatorname{Tr}(X). \quad (100)$$

Therefore, the distance between the allocations of $\mu \mathbf{X}$ and $\mathbf{X}$ is bounded by

$$\|\alpha_{\beta,n}(\mu \mathbf{X}) - \mu \alpha_{\beta,n}(\mathbf{X})\|_\infty \leq C\epsilon \operatorname{Tr}\left(\sum_{i=1}^n X_i\right). \quad (101)$$

It one chooses $C = \operatorname{Tr}(X_1 + \dots + X_n)^{-1}$, then one obtains for all $\epsilon > 0$ the following holds for all sufficiently small μ ,

$$\|\alpha_{\beta,n}(\mu \mathbf{X}) - \mu \alpha_{\beta,n}(\mathbf{X})\|_\infty \leq \epsilon, \quad (102)$$

and therefore the claim holds. $\square$

Theorem 18. *The closure, $\bar{\mathcal{A}}_{\beta,n}$, of the set of all possible allocations of (shifted) unnormalized entropies is a convex cone.*

Proof. This follows [Pip03, Theorem 2.2] and Lemmas 16 and 17. $\square$

It appears that the structure of the cones $\bar{\mathcal{A}}_{\beta,n}$ do not depend on the value of β in any meaningful way and they are all related by an appropriate re-scaling.

Corollary 19. *Let $\beta_1, \beta_2 \in \mathbb{R}$. There exists a bijection from $\mathcal{A}_{\beta_1,n}$ to $\mathcal{A}_{\beta_2,n}$ given by multiplying componentwise by a factor of $\gamma = \exp(\beta_1 - \beta_2)$.*

Proof. The deformation-homogeneous property of H_β (Eq. (70)) implies that for $\gamma := \exp(\beta_1 - \beta_2)$, the following holds

$$H_{\beta_2}(\gamma X) = \gamma H_{\beta_1}(X). \quad (103)$$

By multiplying a given n -tuple $\mathbf{X} = (X_1, \dots, X_n)$ entry-wise by the same factor γ , one concludes that if $\alpha_{\beta_1,n} \in \mathcal{A}_{\beta_1,n}$ then $\exp(\beta_1 - \beta_2) \alpha_{\beta_1,n} \in \mathcal{A}_{\beta_2,n}$. $\square$

Given that each of the convex cones associated to allocations of unnormalized entropies are essentially the same, we will henceforth refer to the **additive quantum entropy cone** as the closure of the set associated to $\beta = 0$:

$$\bar{\mathcal{A}}_n := \bar{\mathcal{A}}_{0,n}. \quad (104)$$

4.2.5 Connection to the multiplicative quantum entropy cone

The purpose of this section is to compare the *additive* quantum entropy cone, $\bar{\mathcal{A}}_{0,n}$, to the (*multiplicative*) quantum entropy cone, $\bar{\mathcal{M}}_n$, previously studied by Pippenger [Pip03].

To begin we prove that the additive quantum entropy cone, $\bar{\mathcal{A}}_{0,n}$, lies inside the multiplicative quantum entropy cone, $\bar{\mathcal{M}}_n$, by following a particular family of quantum states which we now define. defn:spec_embed

Definition 4. Let $(X_1, \dots, X_n)$ be a n -tuple of $d \times d$ PSD matrices. Let $\{|1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^d$ and $\{|0\rangle, |1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^{d+1}$. Fix a subset $S \subset \{1, \dots, n\}$ of size $|S| = k$ and let $X_S := \sum_{i \in S} X_i$. Next define a Hermitian operator, Φ_S , acting on $(\mathbb{C}^{d+1})^{\otimes n}$ with coefficients with respect to the computational bases $|i_1 \dots i_n\rangle$ being given by the following formula:

$$\langle i_1 \dots i_n | \Phi_S | j_1 \dots j_n \rangle = \begin{cases} \langle i_p | X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} | j_q \rangle & \exists! i_p, j_q > 0, p, q \in S \\ 0 & \text{otherwise.} \end{cases} \quad (105)$$

Specifically, $\langle i_1 \cdots i_k | \Phi_S | j_1 \cdots j_n \rangle$ is non-zero only if there exists a unique p such that $i_p > 0$ and a unique q such that $j_q > 0$ and also that such unique indices belong to S , $p, q \in S$. In this case, $\langle i_1 \cdots i_n | \Phi_S | j_1 \cdots j_n \rangle$ takes the value of $\langle i_p | X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} | j_q \rangle$, i.e., the inner product between the i_p th column of $X_p^{\frac{1}{2}}$ and the j_q th column of $X_q^{\frac{1}{2}}$. Up to shuffling the n copies of $\mathbb{C}^{d+1}$ in $(\mathbb{C}^{d+1})^{\otimes n}$, the operator Φ_S is of the form $\tilde{\Phi}_S \otimes Z_{[n] \setminus S}$ where

$$Z_{[n] \setminus S} = |0 \overset{n-k}{\cdots} 0\rangle \langle 0 \overset{n-k}{\cdots} 0|. \quad (106)$$

and where $\tilde{\Phi}_S$ acts merely on $(\mathbb{C}^{d+1})^{\otimes k}$. Also define the operator Γ_S which is $\tilde{\Phi}_S$ shifted by Z_S

$$\Gamma_S := \tilde{\Phi}_S + (1 - \text{Tr}(X_S))Z_S, \quad (107)$$

and in particular $\Gamma_\emptyset := 1$.

Example 2. Given two PSD $d \times d$ matrices, (X_1, X_2) , the matrix $\Gamma := \Gamma_{\{1,2\}}$ has the following coefficients for all $i, j, k, l \in \{1, \dots, d\}$.

$$\langle 00 | \Gamma | 00 \rangle = 1 - \text{Tr}(X_{1,2}), \quad (108)$$

$$\langle i0 | \Gamma | j0 \rangle = \langle i | X_1 | j \rangle, \quad (109)$$

$$\langle 0i | \Gamma | 0j \rangle = \langle i | X_2 | j \rangle, \quad (110)$$

$$\langle i0 | \Gamma | 0j \rangle = \langle i | X_1^{\frac{1}{2}} X_2^{\frac{1}{2}} | j \rangle, \quad (111)$$

$$\langle 0i | \Gamma | j0 \rangle = \langle i | X_2^{\frac{1}{2}} X_1^{\frac{1}{2}} | j \rangle, \quad (112)$$

$$\langle ik | \Gamma | jl \rangle = 0. \quad (113)$$

Example 3. Given three $d \times d$ PSD matrices $A, B, C \geq 0$, the operator $\Gamma := \Gamma_{\{1,2,3\}}$ is the $(d+1)^3 \times (d+1)^3$ matrix with non-zero coefficients given for all $i, j \in \{1, \dots, d\}$ by the formulas:

$$\begin{aligned} \langle 000 | \Gamma | 000 \rangle &= 1 - \text{Tr}(A + B + C), \\ \langle i00 | \Gamma | j00 \rangle &= \langle i | A | j \rangle, \\ \langle 0i0 | \Gamma | 0j0 \rangle &= \langle i | B | j \rangle, \\ \langle 00i | \Gamma | 00j \rangle &= \langle i | C | j \rangle, \\ \langle i00 | \Gamma | 0j0 \rangle &= \langle i | A^{\frac{1}{2}} B^{\frac{1}{2}} | j \rangle, \\ \langle 0i0 | \Gamma | j00 \rangle &= \langle i | B^{\frac{1}{2}} C^{\frac{1}{2}} | j \rangle, \\ \langle i00 | \Gamma | 00j \rangle &= \langle i | A^{\frac{1}{2}} C^{\frac{1}{2}} | j \rangle, \\ \langle 0i0 | \Gamma | j00 \rangle &= \langle i | B^{\frac{1}{2}} A^{\frac{1}{2}} | j \rangle, \\ \langle 00i | \Gamma | j00 \rangle &= \langle i | C^{\frac{1}{2}} A^{\frac{1}{2}} | j \rangle, \\ \langle 00i | \Gamma | 0j0 \rangle &= \langle i | C^{\frac{1}{2}} B^{\frac{1}{2}} | j \rangle. \end{aligned} \quad (114)$$

lem:spec_embed

Lemma 20. Let everything be defined as in Definition 4. The follow statements hold for all subsets $S \subseteq [n]$.

(i) Up to padding by zeros, the eigenvalues of $\tilde{\Phi}_S$, Φ_S and X_S are all identical.

$$\text{spec}(\tilde{\Phi}_X) \simeq \text{spec}(\Phi_X) \simeq \text{spec}(X_X). \quad (115)$$

(ii) The eigenvalues of Γ_S are equal to the eigenvalues of $\tilde{\Phi}_S$ plus the additional value of $1 - \text{Tr}(X_S)$:

$$\text{spec}(\Gamma_S) \simeq \text{spec}(\tilde{\Phi}_S) \oplus \{1 - \text{Tr}(X_S)\}. \quad (116)$$

(iii) The partial trace of $\Gamma_{[n]}$ over $[n] \setminus S$ is equal to Γ_S :

$$\text{Tr}_{[n] \setminus S}(\Gamma_{[n]}) = \Gamma_S. \quad (117)$$

Proof. (i) From the definition of Φ_S , one observes that $\langle i_1 \cdots i_n | \Phi_S | i_1 \cdots i_n \rangle$ is only non-zero if there exists a unique non-zero $i_p \neq 0$ where $p \in S$ and thus

$$\text{Tr}(\Phi_S) = \sum_{p \in S} \sum_{j=1}^d \langle j | X_p | j \rangle = \sum_{p \in S} \text{Tr}(X_p) = \text{Tr}(X_S). \quad (118)$$

More generally, for any positive integer $\ell \in \mathbb{N}_{>0}$, one observes that $\text{Tr}(\Phi_S^\ell) = \text{Tr}(X_S^\ell)$. (**TODO: Prove this directly.**) As the eigenvalues of any matrix X are determined (up to padding by zeros) by the values of $\text{Tr}(X^\ell)$, we conclude that Φ_S and X_S have the same eigenvalues (up to padding by zeros) (**TODO: Is there a more direct way to prove this.**)

(ii) Note that $\tilde{\Phi}_S$ and its eigenvectors are supported in the subspace of $(\mathbb{C}^{d+1})^{\otimes k}$ spanned by $|i_1 \cdots i_k\rangle$ for which exactly one i_p is non-zero. Therefore $|0 \cdots 0\rangle$ is orthogonal to all eigenvectors of $\tilde{\Phi}_S$ and thus $|0 \cdots 0\rangle$ is an eigenvector of Γ_S with eigenvalue $1 - \text{Tr}(X_S)$.

(iii) The partial trace of $\Phi_{[n]}$ over $[n] \setminus S$, when performed in the computational basis $|i_1 \cdots i_n\rangle$, can be divided into three portions depending on the number of non-zero entries in $(i_p \mid p \in [n] \setminus S)$; first, is where all indices i_p are zero, second, where exactly one i_p is non-zero, and third where more than one index i_p is non-zero. The respective contributions are as follows:

$$\text{Tr}_{[n] \setminus S}(\Phi_{[n]}) = \tilde{\Phi}_S + \text{Tr}(X_{[n] \setminus S})Z_S + 0 \quad (119)$$

Recalling the definition of Γ_S , one concludes

$$\begin{aligned} \text{Tr}_{[n] \setminus S}(\Gamma_{[n]}) &= \text{Tr}_{[n] \setminus S}(\Phi_{[n]} + (1 - \text{Tr}(X_{[n]}))Z_{[n]}), \\ &= \tilde{\Phi}_S + \text{Tr}(X_{[n] \setminus S})Z_S + (1 - \text{Tr}(X_{[n]}))Z_S, \\ &= \tilde{\Phi}_S + (1 - \text{Tr}(X_S))Z_S, \\ &= \Gamma_S. \end{aligned} \quad (120)$$

□

Corollary 21. *By Lemma 20, one has $\text{Tr}(\Gamma_S) = 1$. If the eigenvalues of X_S satisfy*

$$\text{Tr}(X_S) \leq 1, \quad (121)$$

then Γ_S is positive semidefinite, and thus Γ_S is a density matrix. In such a case, assuming $X_S \neq 0$ so that $\text{Tr}(X_S) \in (0, 1]$, the von Neumann entropy of Γ_S is related to the von Neumann entropy of $X_S / \text{Tr}(X_S)$ by the equation

$$H_{\text{vN}}(\Gamma_S) = h_{\text{bin}}(\text{Tr}(X_S)) + \text{Tr}(X_S)H_{\text{vN}}\left(\frac{X_S}{\text{Tr}(X_S)}\right). \quad (122)$$

Moreover, if the eigenvalues of X_S satisfy the stronger condition

$$\lambda_1(X_S) + \text{Tr}(X_S) \leq 1, \quad (123)$$

then the maximum eigenvalue of Γ_S is $1 - \text{Tr}(X_S)$.

Remark 7. In order to discover how the additive quantum entropy cone lives inside the multiplicative quantum entropy cone, it suffices to examine how the formula for the von Neumann entropy of the reduced state Γ_S on subsystem S ,

$$H_{\text{vN}}(\Gamma_S) = \text{Tr}(X_S)H_{\text{vN}}\left(\frac{X_S}{\text{Tr}(X_S)}\right) - \text{Tr}(X_S) \log \text{Tr}(X_S) - (1 - \text{Tr}(X_S)) \log(1 - \text{Tr}(X_S)), \quad (124)$$

compares to the formula for the unnormalized entropy of the partial sum $X_S = \sum_{i \in S} X_i$,

$$H_0(X_S) = \text{Tr}(X_S)H_{\text{vN}}\left(\frac{X_S}{\text{Tr}(X_S)}\right) - \text{Tr}(X_S) \log \text{Tr}(X_S). \quad (125)$$

The only term that is different is the additional $(1 - \text{Tr}(X_S)) \log(1 - \text{Tr}(X_S))$ term appearing in $H_{\text{vN}}(\Gamma_S)$:

$$H_0(X_S) = H_{\text{vN}}(\Gamma_S) + (1 - \text{Tr}(X_S)) \log(1 - \text{Tr}(X_S)). \quad (126)$$

In the limit of very small PSD matrices appearing in the n -tuple $(X_1, \dots, X_n)$, this additional term contributes approximately zero to the value of $H_{\text{vN}}(\Gamma_S)$. This can be formalized as follows. If Γ_S is the reduced state obtained via Definition 4 for the n -tuple $\mathbf{X} = (X_1, \dots, X_n)$, then let Γ_S^μ be the reduced state obtained via Definition 4 for the n -tuple $\mu\mathbf{X} = (\mu X_1, \dots, \mu X_n)$.

$$H_{\text{vN}}(\Gamma_S^\mu) = \mu H_{\text{vN}}(\Gamma_S) - \text{Tr}(X_S) \mu \log \mu - (1 - \mu \text{Tr}(X_S)) \log(1 - \mu \text{Tr}(X_S)), \quad (127)$$

$$H_0(\mu X_S) = \mu H_0(X_S) - \text{Tr}(X_S) \mu \log \mu. \quad (128)$$

Now assume that $\mu > 0$ is taken to be small. Now let $\alpha = (\alpha_S)_{S \subseteq n} \in \bar{\mathcal{A}}_n$ be an element of the additive quantum entropy cone. **TODO: Figure out if/how the additive entropy cone lies inside the multiplicative entropy cone.**

5 Asymptotics of $SU(d)$ $6j$ operators

The purpose of this section is to derive some upper and lower bounds on generalized $6j$ symbol for $SU(d)$ in relation to the existence or non-existence of Hermitian operators satisfying the tetrahedral relations Eq. (1). This section essentially constitutes a generalization of the proof techniques in the “quantum-information-theoretic” approach taken in [Chr08]. Specifically, the results of this section are derived from considering a type of pseudo-commutative binomial expansion of tensor powers of sums of Hermitian operators and applying spectrum estimation to the resulting terms.

5.1 Schur-Weyl decomposition & spectrum estimation

Throughout this section, we make use of the following decomposition of a tensor-power of a d -dimensional complex vector space.

Proposition 22 (Schur-Weyl decomposition). *Let n and d be positive integers. Then the decomposition of $(\mathbb{C}^d)^{\otimes n}$ into irreducible representations of $S_n \times SU(d)$, known as the Schur-Weyl decomposition is given by*

$$(\mathbb{C}^d)^{\otimes n} \cong \bigoplus_{\lambda \vdash_n} W_\lambda \otimes V_\lambda^d \quad \text{\textcolor{blue}{\small{eq:schur_weyl_decomp}}} \quad (129)$$

where the direct sum is indexed by Young diagrams λ with n boxes and at most $d - 1$ rows.

The following result constrains the associated Schur-Weyl decomposition of the n -th tensor power, $X^{\otimes n}$, of a positive semidefinite Hermitian matrix X [Chr08; KW01].

Proposition 23 (Spectrum estimation). *Let P_λ denote the Hermitian projection operator associated to the isotypic subspace indexed by $\lambda \vdash_n$ in Eq. (129) and let X be a positive semidefinite matrix with eigenvalues χ . Then*

$$\text{Tr}(P_\lambda X^{\otimes n}) \leq |\chi|^n \dim(V_\lambda^d) \exp(-nH(\bar{\lambda}|\bar{\chi})), \quad (130)$$

where $H(\bar{\lambda}|\bar{\chi})$ denotes the Kullback-Leibler divergence of $\bar{\lambda}$ from $\bar{\chi}$.

Proof. See [KW01] for a proof (noting that the $|\chi|^n$ term simply accounts for the possibility that $\text{Tr}(X) \neq 1$ because $\text{Tr}(X) = |\chi|$). $\square$

5.2 Schur-Weyl presentation of the $6j$ -operator

Lemma 24. *Let x, y, z arbitrary positive integers and let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be Young diagrams of the following sizes:*

$$\begin{array}{lll} \alpha \vdash_d x, & \beta \vdash_d y, & \gamma \vdash_d x + y, \\ \delta \vdash_d z, & \epsilon \vdash_d x + y + z, & \phi \vdash_d y + z, \end{array} \quad \text{\textcolor{blue}{\small{eq:xyz_sized_young_diagrams}}} \quad (131)$$

and define corresponding projection operators acting on $(\mathbb{C}^d)^{\otimes x+y+z}$ by

$$\begin{array}{lll} \Pi_\alpha := P_\alpha \otimes \text{id}^{\otimes y} \otimes \text{id}^{\otimes z}, & \Pi_\beta := \text{id}^{\otimes x} \otimes P_\beta \otimes \text{id}^{\otimes z}, & \Pi_\gamma := P_\gamma \otimes \text{id}^{\otimes z}, \\ \Pi_\delta := \text{id}^{\otimes x} \otimes \text{id}^{\otimes y} \otimes P_\delta, & \Pi_\epsilon := P_\epsilon, & \Pi_\phi := \text{id}^{\otimes x} \otimes P_\phi. \end{array} \quad \text{\textcolor{blue}{\small{eq:extended_projs}}} \quad (132)$$

Then every pair of projection operators in the set $\{\Pi_\alpha, \Pi_\beta, \Pi_\gamma, \Pi_\delta, \Pi_\epsilon, \Pi_\phi\}$ is commuting with only possible exception being Π_γ and Π_ϕ . Then $\Pi_{\alpha\beta\delta\epsilon}^\gamma$ and $\Pi_{\alpha\beta\delta\epsilon}^\phi$, defined by

$$\begin{aligned} \Pi_{\alpha\beta\delta\epsilon}^\gamma &= \Pi_\gamma \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon, \\ \Pi_{\alpha\beta\delta\epsilon}^\phi &= \Pi_\phi \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon. \end{aligned} \quad \text{\textcolor{blue}{\small{eq:commuting_extended_projs}}} \quad (133)$$

are projection operators. Moreover, then $6j$ -operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ has operator norm satisfying

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty = \left\| \Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma \right\|_\infty. \quad \text{\textcolor{blue}{\small{eq:ud_sixj_prod_proj_operator_norm}}} \quad (134)$$

and Hilbert-Schmidt norm satisfying

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\text{HS}}^2 = \frac{\text{Tr}(\Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma)}{\dim(V_\epsilon^d) \dim(W_\alpha) \dim(W_\beta) \dim(W_\delta)}. \quad \text{\textcolor{blue}{(eq:ud_sixj_prod_proj_hs_norm)}} \quad (135)$$

Proof. That all projection operators except the pair (Π_γ, Π_ϕ) commute follows from elementary applications of Schur's lemma; the obstruction for the pair (Π_γ, Π_ϕ) is simply that Π_γ is not necessarily $\text{SU}(d)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes y+z}$ and likewise Π_ϕ is not necessarily $\text{SU}(d)$ -equivariant with respect to the diagonal on $\mathbb{C}^{\otimes x+y}$. To prove both Eq. (134) and Eq. (135), it suffices to show there exists appropriately-typed isometries ι_I and ι_{II} such that

$$\Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma = \iota_{II} \left(\text{id}_{V_\epsilon^d} \otimes \text{id}_{W_\alpha} \otimes \text{id}_{W_\beta} \otimes \text{id}_{W_\delta} \otimes \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right) \iota_I^\dagger \quad \text{\textcolor{blue}{(eq:ud_sixj_prod_proj_iso)}} \quad (136)$$

This follows from the follow diagrammatic calculation TODO: Explain this.:

$$= W_\alpha V_\alpha W_\beta V_\beta W_\delta V_\delta = \dots = V_\epsilon. \quad (137)$$

defn:Q_quantity $\square$

Definition 5. Let (A, B, C, D, E, F) be a tetrahedral sextuple of non-zero positive semidefinite $d \times d$ Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Let x, y, z be arbitrary positive integers and define

$$\Omega := \frac{A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z}}{|\alpha|^x |\beta|^y |\delta|^z}. \quad (138)$$

Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams with sizes determined by Eq. (131), and let $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ and $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ be defined according to Eq. (133). Define $Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \in \mathbb{C}$ by

$$Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} := \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Omega \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}). \quad (139)$$

Proposition 25. Let everything be as in Definition 5. Then

$$Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'} \Omega), \quad \text{\textcolor{blue}{(eq:Q_alt)}} \quad (140)$$

and therefore

$$1 = \text{Tr}(\Omega) = \sum_{\alpha' \vdash_d x} \sum_{\beta' \vdash_d y} \sum_{\gamma' \vdash_d x+y} \sum_{\delta' \vdash_d z} \sum_{\epsilon' \vdash_d y+z} \sum_{\phi' \vdash_d x+y+z} Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \quad \text{\textcolor{blue}{\small{eq:completeness_relation}}} \quad (141)$$

Moreover, the magnitude of $Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'}$ can be upper bounded by either

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq (\text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega) \text{Tr}(\Pi_{\phi'} \Omega))^{\frac{1}{6}}. \quad \text{\textcolor{blue}{\small{eq:Q_pre_upper_bound}}} \quad (142)$$

or

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq \left\| \Pi_{\alpha' \beta' \delta' \epsilon'}^{\phi'} \Pi_{\alpha' \beta' \delta' \epsilon'}^{\gamma'} \right\|_{\infty}. \quad \text{\textcolor{blue}{\small{eq:Q_sixj_upper_bound}}} \quad (143)$$

Proof. The proof of Eq. (140) and Eq. (141) follows from elementary properties of the projection operators defined in Lemma 24. Moreover, Eq. (143) follows because Ω is positive semidefinite and trace one. To prove Eq. (142), consider an application of the Cauchy-Schwarz inequality to the Hilbert-Schmidt inner product, $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$, whenever $X = \Omega^{\frac{1}{2}} \Pi_1$ and $Y = \Omega^{\frac{1}{2}} \Pi_2$ (where Π_1 and Π_2 are generic Hermitian projectors and Ω is positive semidefinite):

$$\begin{aligned} |\text{Tr}(\Pi_1 \Omega \Pi_2)|^2 &= \left| \text{Tr}((\Omega^{\frac{1}{2}} \Pi_1)^\dagger (\Omega^{\frac{1}{2}} \Pi_2)) \right|^2 = \left| \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_2 \rangle \right|^2 \\ &\leq \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_1 \rangle \langle \Omega^{\frac{1}{2}} \Pi_2, \Omega^{\frac{1}{2}} \Pi_2 \rangle = \text{Tr}(\Pi_1 \Omega) \text{Tr}(\Pi_2 \Omega) \end{aligned} \quad \text{\textcolor{blue}{\small{eq:biproj_ineq}}} \quad (144)$$

Letting $\Pi_1 = \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}$ and $\Pi_2 = \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'}$ and applying Eq. (144) yields

$$\begin{aligned} \left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right|^6 &= |\text{Tr}[(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}) \Omega (\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Omega)^3 \text{Tr}(\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi'} \Omega) \text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega). \end{aligned} \quad (145)$$

\text{\textcolor{blue}{\small{thm:Q_upper_bound}}} \quad \square

Theorem 26. Let everything be as in Definition 5 except fix x, y, z to be

$$x := \lceil n |\alpha| \rceil, \quad y := \lceil n |\beta| \rceil, \quad z := \lceil n |\delta| \rceil, \quad (146)$$

where n is a sufficiently large positive integer ensuring that x, y and z are non-zero. Then

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq p(n) \exp(-n I_{\alpha \beta \gamma \delta \epsilon \phi}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} / 6). \quad \text{\textcolor{blue}{\small{eq:Q_upper_bound}}} \quad (147)$$

where $p(n)$ is a polynomial in n and where

$$I_{\alpha \beta \gamma \delta \epsilon \phi}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} = |\alpha| H(\bar{\alpha}' | \bar{\alpha}) + |\beta| H(\bar{\beta}' | \bar{\beta}) + |\gamma| H(\bar{\gamma}' | \bar{\gamma}) + |\delta| H(\bar{\delta}' | \bar{\delta}) + |\epsilon| H(\bar{\epsilon}' | \bar{\epsilon}) + |\phi| H(\bar{\phi}' | \bar{\phi}). \quad (148)$$

TODO: Is this relative entropy deviation measure conceptually useful or is it best to relax the deviation measure to its geometric analogue using Pinsker's inequality?

Proof. To begin the proof, we consider x, y and z to be arbitrary positive integers. For each $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda'} \Omega)$ using the spectrum estimation technique Proposition 23. For a detailed example, we will consider the bound on $\text{Tr}(\Pi_{\gamma'} \Omega)$.

$$\text{Tr}(\Pi_{\gamma'} \Omega) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes z})(A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z})]}{|\alpha|^x |\beta|^y |\delta|^z} = \frac{\text{Tr}[P_{\gamma'}(A^{\otimes x} \otimes B^{\otimes y})] |\delta|^z}{|\alpha|^x |\beta|^y |\delta|^z}. \quad (149)$$

Using the assumption that $C = A + B$, we have $C^{\otimes x+y} = (A + B)^{\otimes x+y}$ which implies

$$\text{Tr}(P_{\gamma'} C^{\otimes x+y}) = \sum_{k=0}^{x+y} \binom{x+y}{k} \text{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes x+y-k})). \quad (150)$$

Here we have used the S_{x+y} -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq k \leq x+y$ that

$$\mathrm{Tr}(P_{\gamma'} C^{\otimes x+y}) \geq \binom{x+y}{k} \mathrm{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes x+y-k})). \quad (151)$$

Letting $k = x$ and rearranging yields

$$\mathrm{Tr}(P_{\gamma'} (A^{\otimes x} \otimes B^{\otimes y})) \leq \frac{x!y!}{(x+y)!} \mathrm{Tr}(P_{\gamma'} C^{\otimes x+y}). \quad (152)$$

Furthermore, spectrum estimation (Proposition 23) yields

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{1}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \mathrm{Tr}[P_{\gamma'} C^{\otimes x+y}], \quad (153)$$

$$\leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^d) \exp(-(x+y)H(\bar{\gamma}'|\bar{\gamma})). \quad (154)$$

Analogous calculations for $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ produce the following set of upper bounds:

$$\mathrm{Tr}(\Pi_{\alpha'} \Omega) \leq \dim(V_{\alpha'}^d) \exp(-xH(\bar{\alpha}'|\bar{\alpha})), \quad (155)$$

$$\mathrm{Tr}(\Pi_{\beta'} \Omega) \leq \dim(V_{\beta'}^d) \exp(-yH(\bar{\beta}'|\bar{\beta})), \quad (156)$$

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^d) \exp(-(x+y)H(\bar{\gamma}'|\bar{\gamma})), \quad (157)$$

$$\mathrm{Tr}(\Pi_{\delta'} \Omega) \leq \dim(V_{\delta'}^d) \exp(-zH(\bar{\delta}'|\bar{\delta})), \quad (158)$$

$$\mathrm{Tr}(\Pi_{\epsilon'} \Omega) \leq \frac{(|\alpha| + |\beta| + |\delta|)^{x+y+z}}{|\alpha|^x |\beta|^y |\delta|^z} \frac{x!y!z!}{(x+y+z)!} \dim(V_{\epsilon'}^d) \exp(-(x+y+z)H(\bar{\epsilon}'|\bar{\epsilon})), \quad (159)$$

$$\mathrm{Tr}(\Pi_{\phi'} \Omega) \leq \frac{(|\beta| + |\delta|)^{x+y+z}}{|\beta|^y |\delta|^z} \frac{y!z!}{(y+z)!} \dim(V_{\phi'}^d) \exp(-(y+z)H(\bar{\phi}'|\bar{\phi})). \quad (160)$$

Our objective now will be to use the given values for x, y, z to prove the multinomial prefactors appearing in the bounds for γ', ϵ' , and ϕ' reasonably small compared to n . An application of Lemma 35 (when $k = 2$) implies the following bound

$$\frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \leq e^4 (n(|\alpha| + |\beta|) + 3)^2. \quad (161)$$

Moreover, using [CM06, Equation 7], we have

$$\dim(V_{\gamma'}^d) \leq (|\gamma'| + 1)^{d(d-1)/2} = (x+y+1)^{d(d-1)/2} < (n(|\alpha| + |\beta|) + 3)^{d(d-1)/2}. \quad (162)$$

Altogether, we can relax the bound on $\mathrm{Tr}(\Pi_{\gamma'} \Omega)$ to obtain

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq e^4 (n|\epsilon| + 3)^{\frac{d(d-1)}{2}+2} \exp(-n|\gamma| H(\bar{\gamma}'|\bar{\gamma})). \quad (163)$$

Similar calculations for $\mathrm{Tr}(\Pi_{\epsilon'} \Omega)$ yields (noting that $k = 3$ when applying Lemma 35):

$$\mathrm{Tr}(\Pi_{\epsilon'} \Omega) \leq e^{16} (n|\epsilon| + 4)^{\frac{d(d-1)}{2}+3} \exp(-n|\epsilon| H(\bar{\epsilon}'|\bar{\epsilon})). \quad (164)$$

It is also not too difficult to see that the polynomial in n term associated to the upper bound on $\mathrm{Tr}(\Pi_{\epsilon'} \Omega)$ above is the largest such subexponential term and thus Eq. (147) indeed holds if we take $p(n)$ to be

$$p(n) = e^{16} (n|\epsilon| + 4)^{\frac{d(d-1)}{2}+3}. \quad (165)$$

□

Corollary 27. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) . Then there exists a sextuple of sequences of Young diagrams $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ of sizes

$$\begin{aligned} \alpha'_n \vdash_d [n|\alpha|], & \quad \beta'_n \vdash_d [n|\beta|], & \quad \gamma'_n \vdash_d [n|\alpha|] + [n|\beta|], \\ \delta'_n \vdash_d [n|\delta|], & \quad \epsilon'_n \vdash_d [n|\alpha|] + [n|\beta|] + [n|\delta|], & \quad \phi'_n \vdash_d [n|\beta|] + [n|\delta|], \end{aligned} \quad (166)$$

satisfying (i) the convergence conditions

$$\begin{aligned} \lim_{n \rightarrow \infty} \bar{\alpha}'_n &= \bar{\alpha}, & \lim_{n \rightarrow \infty} \bar{\beta}'_n &= \bar{\beta}, & \lim_{n \rightarrow \infty} \bar{\gamma}'_n &= \bar{\gamma}, \\ \lim_{n \rightarrow \infty} \bar{\delta}'_n &= \bar{\delta}, & \lim_{n \rightarrow \infty} \bar{\epsilon}'_n &= \bar{\epsilon}, & \lim_{n \rightarrow \infty} \bar{\phi}'_n &= \bar{\phi}, \end{aligned} \quad (167)$$

and (ii) the lower bound

$$\left\| \left\{ \frac{\alpha'_n}{\delta'_n} \frac{\beta'_n}{\epsilon'_n} \frac{\gamma'_n}{\phi'_n} \right\} \right\|_{\infty} \geq \frac{1}{(n|\epsilon| + 4)^{6(d-1)}}. \quad (168)$$

Proof. For each n , it suffices to pick $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ be the sextuple which maximizes the value of $\left| Q_{A,B,C,D,E,F}^{\alpha'_n \beta'_n \gamma'_n \delta'_n \epsilon'_n \phi'_n} \right|$. Such a choice will necessarily satisfy a lower bound due to the completeness relation Eq. (141). Specifically,

$$\left| Q_{A,B,C,D,E,F}^{\alpha'_n \beta'_n \gamma'_n \delta'_n \epsilon'_n \phi'_n} \right| \geq \frac{1}{q(n)} \quad (169)$$

where $q(n)$ is any upper bound on the number of possible summands in Eq. (141) viewed as a function of n . As there are at most $(k+1)^{d-1}$ Young diagrams with $d-1$ rows and size at most k , letting

$$q(n) = (n|\epsilon| + 4)^{6(d-1)} \quad (170)$$

ensures Eq. (169) holds because the size of ϵ'_n is at most $k = n|\epsilon| + 3$. Moreover, applying Eq. (147) yields

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_n \beta'_n \gamma'_n \delta'_n \epsilon'_n \phi'_n} \leq \frac{6}{n} (\log q(n) + \log p(n)) \xrightarrow{n \rightarrow \infty} 0 \quad (171)$$

and thus Eq. (167) holds because $H(\bar{\lambda}'|\bar{\lambda}) \geq 0$ with equality if and only if $\bar{\lambda}' = \bar{\lambda}$. Then Eq. (168) follows from Eq. (169) combined with Eq. (143) and Eq. (134). $\square$

Theorem 28. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be non-negative non-zero eigenvalues which are not tetrahedral. Then for all sequences of Young diagrams $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ satisfying Eq. (166) and Eq. (167), we have

$$\left\| \left\{ \frac{\alpha'_n}{\delta'_n} \frac{\beta'_n}{\epsilon'_n} \frac{\gamma'_n}{\phi'_n} \right\} \right\|_{\text{HS}}^2 \leq \exp(-n I_*^{\alpha\beta\gamma\delta\epsilon\phi} / 6 + O(\log n)). \quad (172)$$

where $I_*^{\alpha\beta\gamma\delta\epsilon\phi} > 0$ quantifies the deviation of $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ from being tetrahedral:

$$I_*^{\alpha\beta\gamma\delta\epsilon\phi} = \min_{\substack{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi} \\ \text{tetrahedral}}} I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha\beta\gamma\delta\epsilon\phi}. \quad (173)$$

Proof. Let dX be the normalized Hilbert-Schmidt measure over $d \times d$ density matrices and let k be a positive integer. Then define an operator Γ_k on $(\mathbb{C}^d)^{\otimes k}$ by

$$\Gamma_k := \int dX X^{\otimes k}. \quad (174)$$

As Γ_k is known to be both S_k and $\text{SU}(d)$ equivariant, by Schur's lemma, Γ_k can be expressed as

$$\Gamma_k = \sum_{\lambda' \vdash_d k} r_{\lambda'} P_{\lambda'} \quad (175)$$

with coefficients $r_{\lambda'}$ that can be computed [ZS01; Wal14] to obtain

$$r_{\lambda'} = \frac{\dim(V_{\lambda'}^d)}{\dim(W_{\lambda'}) \dim(V_{(k)}^{d^2})}. \quad (176)$$

Now let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be arbitrary Young diagrams with respective sizes $(x, y, x+y, z, x+y+z, y+z)$ where x, y, z are determined by a positive integer n as in Theorem 26. Using Eq. (135), we obtain the formula

$$\left\| \begin{Bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{Bmatrix} \right\|_{\text{HS}}^2 = \frac{\text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'})}{\dim(V_{\epsilon'}^d) \dim(W_{\alpha'}) \dim(W_{\beta'}) \dim(W_{\delta'})} \quad (177)$$

$$= R_{\alpha', \beta', \delta', \epsilon'} \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\Gamma_x \otimes \Gamma_y \otimes \Gamma_z) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (178)$$

where the coefficient $R_{\alpha', \beta', \delta', \epsilon'}$ is a ratio of dimensions:

$$R_{\alpha', \beta', \delta', \epsilon'} := \frac{\dim(V_{(x)}^{d^2}) \dim(V_{(y)}^{d^2}) \dim(V_{(z)}^{d^2})}{\dim(V_{\epsilon'}^d) \dim(V_{\alpha'}^d) \dim(V_{\beta'}^d) \dim(V_{\delta'}^d)}. \quad (179)$$

Moreover, by the definition of Γ_k , we have

$$\text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\Gamma_x \otimes \Gamma_y \otimes \Gamma_z) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (180)$$

$$= \int dX \int dY \int dZ \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (X^{\otimes x} \otimes Y^{\otimes y} \otimes Z^{\otimes z}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}), \quad (181)$$

where the integral is taken over a triple of $d \times d$ density matrices X, Y, Z . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha| X, & \tilde{B} &:= |\beta| Y, & \tilde{C} &:= |\alpha| X + |\beta| Y, \\ \tilde{D} &:= |\delta| Z, & \tilde{E} &:= |\alpha| X + |\beta| Y + |\gamma| Z, & \tilde{F} &:= |\beta| Y + |\delta| Z. \end{aligned} \quad (182)$$

Furthermore, let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is equal to

$$Q_{\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} = \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (X^{\otimes x} \otimes Y^{\otimes y} \otimes Z^{\otimes z}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (183)$$

Moreover, by Eq. (143) we have

$$\left| Q_{\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq p(n) \exp(-n I_{\tilde{\alpha} \tilde{\beta} \tilde{\gamma} \tilde{\delta} \tilde{\epsilon} \tilde{\phi}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} / 6). \quad (184)$$

Under the assumption that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral, we conclude that

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \neq (\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi}) \quad (185)$$

and therefore $I_{\tilde{\alpha} \tilde{\beta} \tilde{\gamma} \tilde{\delta} \tilde{\epsilon} \tilde{\phi}}^{\alpha \beta \gamma \delta \epsilon \phi} > 0$. **TODO: Figure out how to describe the convergence, is there a triangle inequality for relative entropies? Maybe we need to relax the rate function I to its geometric analogue using Pinsker's inequality, then everything works.** $\square$

5.3 The asymptotics of irrep dimensions

It will be useful for what follows to better understand how the dimension, $\dim(V_{\lambda}^d)$, of the irreducible representation V_{λ}^d of $\text{SL}(d)$ grows as d becomes much larger $|\lambda|$ while $|\lambda|$ itself is much larger than $\ell(\lambda)$. Itzykson and Nauenberg provide the following estimate for $\dim(V_{\lambda}^d)$ [IN66].

Lemma 29. *Let λ be an positive integer partition of length $\ell := \ell(\lambda)$, let $d \geq \ell$ be a positive integer and let $\rho^d := (d-1, d-2, \dots, 2, 1)$ be the Weyl-vector for $\text{SL}(d)$. The following approximation holds when $d \gg |\lambda|^2$ and $\lambda_{\ell} \gg \ell$:*

$$\dim(V_{\lambda}^d) = \dim(V_{\lambda}^{\ell}) \times \prod_{i=1}^{\ell} \frac{\rho_i^{\ell}!}{(\lambda_i + \rho_i^{\ell})!} \times d^{|\lambda|} \times \left(1 + O\left(\frac{|\lambda|^2}{d}\right) \right). \quad (186)$$

Proof. Weyl's character formula can be used to derive the following formula for $\dim(V_\lambda^d)$ in terms of the Young diagram λ and the dimension d [IN66]:

$$\dim(V_\lambda^d) = \frac{\Delta(\lambda + \rho^d)}{\Delta(\rho^d)} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i + \rho_i^d) - (\lambda_j + \rho_j^d)}{\rho_i^d - \rho_j^d} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i}, \quad (187)$$

where $\Delta(x) = \prod_{i < j} (x_i - x_j)$ is the Vandermonde determinant. Isolating the terms above which depend on d from those that do not yields

$$\dim(V_\lambda^d) = \dim(V_\lambda^\ell) \times \prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \times \prod_{i=1}^{\ell} \frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!}. \quad \text{\textcolor{blue}{\code{eq:dim_separation_large_d}}} \quad (188)$$

Given positive integers a and b such that $b \gg a^2$, one obtains the following approximation:

$$\begin{aligned} \log \left(\frac{(a+b)!}{b!} \right) &= \log \prod_{j=1}^a (b+j) = \sum_{j=1}^a \log(b+j) \\ &= a \log b + \sum_{j=1}^a \log \left(1 + \frac{j}{b} \right) = a \log b + O \left(\frac{a^2}{b} \right). \end{aligned} \quad (189)$$

The assumption that $d \gg |\lambda|^2$ and $i \leq \ell \ll \lambda_\ell \leq |\lambda|$ also implies $d - i = \rho_i^d \gg \lambda_i^2$. Therefore,

$$\log \left(\frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!} \right) = \lambda_i \log \rho_i^d + O \left(\frac{\lambda_i^2}{\rho_i^d} \right). \quad (190)$$

The assumptions also imply the following simplification

$$\log \left(\frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!} \right) = \lambda_i \log d + O \left(\frac{|\lambda|^2}{d} \right) \quad (191)$$

which, when applied to Eq. (188), proves the claim. $\square$

Using Lemma 29, it can be seen that for fixed λ , $\dim(V_\lambda^d)$ generally grows polynomially in d with leading exponent $|\lambda|$ and leading coefficient depending only on λ . The objective of the following lemma is to characterize the asymptotics of this leading coefficient for large $|\lambda|$ and even large d . TODO: Figure out a elegant way to present these “double asymptotics”, maybe just epsilonfying everything, like $\max\{\ell/|\lambda|, |\lambda|/d\} \leq \epsilon$. \code{lem:double_asymptotics}

Lemma 30. *Let everything be as in Lemma 29. Then*

$$\log \dim(V_\lambda^d) = |\lambda| + |\lambda| \log(d) + s(\lambda) + O(\ell^2 \log |\lambda|) + O \left(\frac{|\lambda|^2}{d} \right) \quad (192)$$

where $s(\lambda)$ is defined by

$$s(\lambda) := - \sum_{i=1}^{\ell(\lambda)} \lambda_i \log \lambda_i. \quad \text{\textcolor{blue}{\code{eq:unnorm_shannon}}} \quad (193)$$

Proof. The proof proceeds by considering the multiplicands in Eq. (186) separately. First, we upper bound the dimension $\dim(V_\lambda^\ell)$; for indices $i < j \leq \ell$, one has $(\lambda_i - i) - (\lambda_j - j) \leq |\lambda| + 1$ as well as $j - i \geq 1$ which implies

$$\dim(V_\lambda^\ell) = \prod_{1 \leq i < j \leq \ell} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i} \leq (|\lambda| + 1)^{\ell(\ell-1)/2}, \quad \text{\textcolor{blue}{\code{eq:dim_lambda_ell_bound}}} \quad (194)$$

and thus $\log \dim(V_\lambda^\ell) = O(\ell^2 \log |\lambda|)$.

Second, we apply Stirling's approximation [Rob55] to the second term in Eq. (186) under the given assumption that $\lambda_\ell \gg \ell$ (which implies $\rho_i^\ell \ll \lambda_i$),

$$\log \left(\prod_{i=1}^{\ell} \frac{\rho_i^{\ell}!}{(\lambda_i + \rho_i^{\ell})!} \right) = \sum_{i=1}^{\ell} \log \rho_i^{\ell}! - \sum_{i=1}^{\ell} \log(\lambda_i + \rho_i^{\ell})!, \quad (195)$$

$$= |\lambda| + \sum_{i=1}^{\ell} \left(\rho_i^{\ell} \log \rho_i^{\ell} - (\lambda_i + \rho_i^{\ell}) \log(\lambda_i + \rho_i^{\ell}) + O(\log \lambda_i) \right), \quad (196)$$

$$= |\lambda| - \sum_{i=1}^{\ell} \lambda_i \log \lambda_i + O(\ell \log |\lambda|), \quad (197)$$

$$\text{\textcolor{blue}{(eq:unnorm_shannon_term)}} \quad (198)$$

Combining Eq. (194) and Eq. (198) with Lemma 29 proves the claim. $\square$

Remark 8. Note that the quantity $s(\lambda)$ defined in Eq. (193) resembles the Shannon entropy,

$$S(p) = - \sum_{i=1}^{\ell} p_i \log p_i \quad (199)$$

of a probability distribution $p = (p_1, \dots, p_\ell)$. Indeed, $s(\lambda)$ is related to the Shannon entropy of the *normalization* of λ by the formula

$$s(\lambda) = |\lambda| S \left(\frac{\lambda}{|\lambda|} \right) - |\lambda| \log |\lambda|. \quad \text{\textcolor{blue}{(eq:little_s_relation)}} \quad (200)$$

TODO: Perhaps a more direct way of relating the asymptotics of $\dim(V_\lambda^d)$ to $\dim(W_\lambda)$ can be found in Harrow's paper [Har23, Lemma 2]:

Lemma 31. *Let λ be a Young diagram with at most d parts. Then*

$$\frac{|\lambda|!}{d^{|\lambda|}} \frac{\dim(V_\lambda^d)}{\dim(W_\lambda)} = \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{d} \right) \quad (201)$$

5.4 An entropic inequality for tetrahedral spectra

thm:ud_recouple_bound

Theorem 32. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in \mathbb{Z}_{\text{dec}}^d$ be a sextuple of dominant weights for $\text{GL}(d)$ and let $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ be the corresponding $\text{GL}(d)$ -recoupling operator. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{HS}^2 \leq \left(c_{\alpha, \phi}^{\epsilon} c_{\beta, \delta}^{\phi} c_{\alpha, \beta}^{\gamma} c_{\gamma, \delta}^{\epsilon} \right)^{\frac{1}{2}} \frac{\dim(V_\gamma) \dim(V_\phi)}{\dim(V_\beta) \dim(V_\epsilon)} \quad (202)$$

Proof. **TODO:** Explain what this all means carefully. Applying both the Cauchy-Schwarz inequality with

respect to the contraction annotated below, and Schur's lemma multiple times yields the following inequality:

$$\begin{aligned}
& \left(\epsilon \right)^2 C_{\alpha, \beta}^{\gamma} C_{\alpha, \phi}^{\epsilon} C_{\beta, \delta}^{\phi} C_{\gamma, \delta}^{\epsilon} = \\
& \left[\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} \right]^{\frac{1}{2}} = \frac{1}{\left(\beta \right) \left(\epsilon \right)} \left[\begin{array}{c} \text{Diagram 3} \\ \text{Diagram 4} \end{array} \right]^{\frac{1}{2}} \\
& = \frac{\left(\epsilon \right)^2 \left(\gamma \right) \left(\phi \right)}{\left(\beta \right) \left(\epsilon \right)} \left[C_{\alpha, \phi}^{\epsilon} C_{\beta, \delta}^{\phi} C_{\alpha, \beta}^{\gamma} C_{\gamma, \delta}^{\epsilon} \right]^{\frac{1}{2}}
\end{aligned} \tag{203}$$

lem:asymptotic_dimension_ratio $\square$

Lemma 33. Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that

$$|\gamma| + |\phi| = |\beta| + |\epsilon| \tag{204}$$

and that $|\epsilon| = \max\{|\gamma|, |\phi|, |\beta|, |\epsilon|\}$. Let $\ell := \max\{\ell(\gamma), \ell(\phi), \ell(\beta), \ell(\epsilon)\}$ and let d be a positive integer such that $d \geq \ell$. Then

$$\log \left(\frac{\dim(V_{\gamma}^d) \dim(V_{\phi}^d)}{\dim(V_{\beta}^d) \dim(V_{\epsilon}^d)} \right) = I(\gamma, \phi; \beta, \epsilon) + O(\ell^2 \log |\epsilon|) + O \left(\frac{|\epsilon|^2}{d} \right) \tag{205}$$

where $I(\gamma, \phi; \beta, \epsilon)$ is the function

$$I(\gamma, \phi; \beta, \epsilon) := s(\gamma) + s(\phi) - s(\beta) - s(\epsilon). \tag{206}$$

Moreover, $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ for any $k > 0$.

Proof. The proof follows from four applications of Lemma 30. The assumption that the sizes of the Young diagrams are balanced in the sense of Eq. (204) implies that the $\log(d)$ terms appearing in Lemma 30 cancel each other and one obtains Eq. (205).

Four applications of Eq. (200) yields

$$\begin{aligned}
I(\gamma, \phi; \beta, \epsilon) &= s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \\
&= \{|\gamma| S(\bar{\gamma}) + |\phi| S(\bar{\phi}) - |\beta| S(\bar{\beta}) - |\epsilon| S(\bar{\epsilon})\} + \dots \\
&\quad \dots - \{|\gamma| \log |\gamma| + |\phi| \log |\phi| - |\beta| \log |\beta| - |\epsilon| \log |\epsilon|\}.
\end{aligned} \tag{207}$$

Under scaling $\lambda \mapsto k\lambda$ for any $k > 0$, observe that $\overline{k\lambda} = \bar{\lambda}$ and $|k\lambda| = k|\lambda|$. Therefore the above expression implies

$$I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon) + (|\gamma| + |\phi| - |\beta| - |\epsilon|)k \log k. \tag{208}$$

Since Eq. (204) is assumed to hold, the coefficient in front of the $k \log k$ term appearing in Eq. (208) vanishes and therefore $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ as claimed. $\square$

Theorem 34. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (1). Then necessarily $|\beta| + |\epsilon| = |\gamma| + |\phi|$ and furthermore*

$$I(\gamma, \phi; \beta, \epsilon) \geq 0. \tag{209}$$

Proof. The proof follows from asymptotics given by Lemma 33 in the limit as $d \rightarrow \infty$, the upper bound provided by Theorem 32, and the lower bound provided by Corollary 27. **TODO: Finish this** $\square$

6 Conclusion

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A Entropic identities & inequalities

In this section, we recall a handful of identities and inequalities concerning multinomial coefficients and the multinomial distribution in the context of Sanov's theorem [DZ10, Chapter 2]. Let $q = (q_1, \dots, q_k)$ be a probability distribution such that for all $i \in \{1, \dots, k\}$, nq_i is an integer. Let $M_n(q)$ denote the following multinomial coefficient:

$$M_n(q) := \binom{n}{nq_1, \dots, nq_k} = \frac{n!}{(nq_1)! \dots (nq_k)!}, \quad \text{\texttt{eq:multinomial_coefficient}} \quad (210)$$

which can be bounded (see [DZ10, Lemma 2.1.8]) in terms of $H(q)$ by ,

$$(n+1)^{-k} e^{nH(q)} \leq M_n(q) \leq e^{nH(q)}, \quad \text{\texttt{eq:multinomial_coefficient_bound}} \quad (211)$$

where $H(q) = -\sum_{i=1}^k q_i \log q_i$ is the Shannon entropy of q . Additionally, we have for any probability distribution $p = (p_1, \dots, p_k)$,

$$p_1^{nq_1} \dots p_k^{nq_k} = e^{-n(H(q) + H(q|p))} \quad \text{\texttt{eq:multinomial_probability}} \quad (212)$$

where $H(q|p) = \sum_{i=1}^k q_i \log q_i/p_i$ denotes the Kullback-Leibler divergence or relative entropy of q from p with the convention that $H(q|p) = \infty$ if $q_i > 0$ whenever $p_i = 0$. Note that $H(q|p) = 0$ if and only if $p = q$. Combining Eq. (211) with Eq. (212) places bounds on the multinomial distribution

$$(n+1)^{-k} e^{-nH(q|p)} \leq \binom{n}{nq_1, \dots, nq_k} p_1^{nq_1} \dots p_k^{nq_k} \leq e^{-nH(q|p)} \quad (213)$$

If needed, it is possible to relax these bounds further using Pinsker's inequality for $H(q|p)$ in conjunction with its reverse (which only holds for $H(q|p) < \infty$) [GSS19]:

$$2d_{\text{TV}}^2(p, q) \leq H(q|p) \leq \frac{4}{p_{\min}} d_{\text{TV}}^2(p, q) \quad (214)$$

where

$$d_{\text{TV}}(p, q) = \max_{1 \leq j \leq k} |p_j - q_j| = \frac{1}{2} \sum_{j=1}^k |p_j - q_j| \quad \text{\texttt{eq:pinskera}} \quad (215)$$

is known as the *total variation distance* between p and q . Now for any distribution $p = (p_1, \dots, p_k)$ and any positive integer n , it is possible to find another distribution $p' = (p'_1, \dots, p'_k)$ such that (i) np'_i is an integer for all $i \in \{1, \dots, k\}$, (ii) $H(p'|p) < \infty$ is finite, and (iii) p' approximates p in the sense that for some constant $C \geq 0$,

$$d_{\text{TV}}(p, p') \leq \frac{C}{n}. \quad \text{\texttt{eq:tv_approx}} \quad (216)$$

Concretely, one can choose p' such that

$$np'_j := \text{nint} \left(n \sum_{i=1}^j p_i \right) - \text{nint} \left(n \sum_{i=1}^{j-1} p_i \right) \quad \text{\texttt{eq:nint_approx}} \quad (217)$$

where $\text{nint} : \mathbb{R} \rightarrow \mathbb{Z}$ is the function which rounds each real number to the nearest integer (with half integers always rounded up). Using this approximation, we have $|np_j - np'_j| \leq 1$ because each rounding operation in Eq. (217) introduces an error of at most $1/2$ and thus Eq. (216) holds with $C = 1$. In any case, when Eq. (216) holds for some constant $C \geq 0$, Eq. (215) reveals $H(p'|p) \leq \frac{4C}{p_{\min} n^2}$ which yields a useful, albeit crude, lower bound for the multinomial distribution:

$$\binom{n}{np'_1, \dots, np'_k} p_1^{np'_1} \dots p_k^{np'_k} \geq \exp \left(-\frac{4C^2}{p_{\min} n} - k \log(n+1) \right). \quad \text{\texttt{lem:multinomial_dist_lower_bound}} \quad (218)$$

Lemma 35. Let $k \geq 2$ and let $(w_1, \dots, w_k) \in \mathbb{R}_{>0}^k$ have minimum $w_{\min} := \min\{w_1, \dots, w_k\} > 0$ and let n be a positive integer satisfying $n \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_k) \in \mathbb{N}_{>0}^k$ by the equation

$$z_i := \lceil nw_i \rceil. \quad (219)$$

Let $Z = z_1 + \dots + z_k$ and $W = w_1 + \dots + w_k$ and defined probability distributions $\bar{z}$ and $\bar{w}$ by

$$\bar{z}_i = \frac{z_i}{Z}, \quad \text{and} \quad \bar{w}_i = \frac{w_i}{W}, \quad (220)$$

for all $i \in \{1, \dots, k\}$. Then

$$\frac{Z!}{z_1! \dots z_k!} \frac{w_1^{z_1} \dots w_k^{z_k}}{W^Z} \geq e^{-4(k-1)^2} (nW + k + 1)^{-k} \quad \text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound}}} \quad (221)$$

Proof. Let $\delta_j := z_j - nw_j$ denote the rounding surplus such that $0 \leq \delta_j < 1$. Note that for $n \gg k$, we have $Z \approx nW$ because

$$nW \leq Z < nW + k. \quad (222)$$

In this regime, we wish to show that $\bar{z}_i \approx \bar{w}_i$.

$$|\bar{z}_i - \bar{w}_i| = \left| \frac{z_i}{Z} - \frac{w_i}{W} \right| = \frac{|Wz_i - Zw_i|}{ZW}. \quad (223)$$

The numerator above can be bounded independently of n by,

$$|Wz_i - Zw_i| \quad (224)$$

$$= |(nw_i + \delta_i)(w_1 + \dots + w_k) - w_i(\delta_1 + nw_1 + \dots + \delta_k + nw_k)|, \quad (225)$$

$$= |\delta_i(w_1 + \dots + \hat{w}_i + \dots + w_k) - w_i(\delta_1 + \dots + \hat{\delta}_i + \dots + \delta_k)|, \quad (226)$$

$$\leq |\delta_i(w_1 + \dots + \hat{w}_i + \dots + w_k)| + |w_i(\delta_1 + \dots + \hat{\delta}_i + \dots + \delta_k)|, \quad (227)$$

$$\leq (w_1 + \dots + \hat{w}_i + \dots + w_k) + (k-1)(w_i), \quad (228)$$

$$\leq (k-1)(w_1 + \dots + w_k), \quad (229)$$

$$= (k-1)W. \quad (230)$$

Consequently,

$$d_{\text{TV}}(\bar{z}, \bar{w}) = \max_{1 \leq i \leq k} |\bar{z}_i - \bar{w}_i| \leq \frac{k-1}{Z} \quad (231)$$

and therefore

$$\frac{Z!}{z_1! \dots z_k!} \frac{w_1^{z_1} \dots w_k^{z_k}}{W^Z} \geq (Z+1)^{-k} \exp(-ZH(\bar{z}|\bar{w})), \quad (232)$$

$$\geq (Z+1)^{-k} \exp \left[-\frac{4ZW}{w_{\min}} \left(\frac{k-1}{Z} \right)^2 \right], \quad (233)$$

$$\geq (nW + k + 1)^{-k} \exp \left[-\frac{4(k-1)^2}{nw_{\min}} \right], \quad (234)$$

$$\geq (nW + k + 1)^{-k} \exp(-4(k-1)^2). \quad (235)$$

□

TODO: Figure out the best lower bound on the multinomial distribution to use here.

\textcolor{blue}{\code{lem:multinomial_dist_lower_bound_alt}}

Lemma 36. Let everything be as in Lemma 35. Then

$$\frac{Z!}{z_1! \dots z_k!} \frac{w_1^{z_1} \dots w_k^{z_k}}{W^Z} \geq \left(\frac{1}{20} \right)^k \left(\frac{W}{w_1 \dots w_k} \right)^{\frac{1}{2}} \frac{\text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound_alt}}}}{n^{\frac{3k-1}{2}}} \quad (236)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$n! = \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n} e^{r_n}, \quad \text{where} \quad \frac{1}{12n+1} \leq r_n \leq \frac{1}{12n}. \quad (237)$$

Therefore,

$$\begin{aligned} \frac{Z!}{z_1! \cdots z_k!} \frac{w_1^{z_1} \cdots w_k^{z_k}}{W^Z} &= \frac{Z!}{W^Z} \frac{w_1^{z_1}}{z_1!} \cdots \frac{w_k^{z_k}}{z_k!} \\ &= \left(\frac{1}{2\pi}\right)^{\frac{k-1}{2}} \left(\frac{W}{w_1 \cdots w_k}\right)^{\frac{1}{2}} \left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \left(\frac{w_1}{z_1}\right)^{z_1+\frac{1}{2}} \cdots \left(\frac{w_k}{z_k}\right)^{z_k+\frac{1}{2}} \exp\left(r_Z - \sum_i r_{z_i}\right). \end{aligned} \quad (238)$$

In this setting, we have assumed $nw_i \geq 1$ which implies

$$nw_i \leq z_i < nw_i + 1 \leq 2nw_i, \quad \text{and} \quad nW \leq Z < nW + k < 2nW. \quad (240)$$

Consequently³,

$$\left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \geq n^{nW+\frac{1}{2}}, \quad \left(\frac{w_i}{z_i}\right)^{z_i+\frac{1}{2}} \geq \left(\frac{w_i}{nw_i+1}\right)^{nw_i+\frac{3}{2}} \geq \frac{2^{-\frac{5}{2}}}{n^{nw_i+\frac{3}{2}}}. \quad (241)$$

and the remainder term satisfies

$$r_Z - \sum_i r_{z_i} \geq \frac{1}{12Z+1} - \sum_i \frac{1}{12z_i} \geq -\frac{k}{12}. \quad (242)$$

Altogether, we have

$$\frac{Z!}{z_1! \cdots z_k!} \frac{w_1^{z_1} \cdots w_k^{z_k}}{W^Z} = \left(\frac{1}{2\pi}\right)^{\frac{k-1}{2}} \left(\frac{W}{w_1 \cdots w_k}\right)^{\frac{1}{2}} \frac{2^{-\frac{5k}{2}}}{n^{\frac{3k-1}{2}}} \exp\left(-\frac{k}{12}\right). \quad (243)$$

Moreover,

$$\left(\frac{1}{2\pi}\right)^{\frac{k-1}{2}} 2^{-\frac{5k}{2}} \exp\left(-\frac{k}{12}\right) \geq \left(\frac{1}{20}\right)^k \quad (244)$$

□

³Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1}\right)^{x+\frac{3}{2}} \geq 2^{-5/2}$

B [Research Notes]

B.1 Horn’s problem & generalizations

- Horn’s original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- An excellent expository overview of the big picture by Knutson [Knu00]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- Quantum-info style proof of Littlewood-Richardson coefficient connection due to Christandl [Chr08]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn’s inequalities in characterizing the existence of invariant vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]
- Fulton provides a comprehensive treatment of the connection between representation theory of $GL(d)$ and the geometry and (co)homology of the complex Grassmannians [Ful97]
- Prakash Belkale has contributed greatly to the theory surrounding Horn-type problems: for example, in [Bel04], Belkale provides a *geometric* proof of the equivalence between (i) the structure coefficients of the representation theory of $GL(d)$ (i.e. the LR-coefficients $c_{\mu\nu}^\lambda$), and (ii) the structure coefficients of the cohomology ring of the Grassmannian $Gr(d, n)$ of d -dimensional subspaces in $\mathbb{C}^d$. This proof is now complementary to the previously known algebraic proofs which rely on knowledge of the algebra of symmetric functions. This geometric approach taken by Belkale, which emphasizes the tangent spaces of the involved Schubert varieties, is later used by Belkale to provide a geometric proof of the saturation conjecture [Bel02].

B.2 Thompson’s conjecture

- Original (indirect) proof through the lense of random walks Klyachko [Kly00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]

- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko
- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).

B.3 $6j$ symbols

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- The masters essay/thesis of Backens (**TODO: check if the pdf Matthias sent to me is the final version**) [Bac10]
- The spectral quantum marginal problem and the $6j$ symbol for the symmetric group Christandl, Şahinoğlu, and Walter [CSW18]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Keppeler, Plätzer, and Sjodahl recently calculated $SU(d)$ $6j$ symbols with adjoint representations [KPS24] and a bit earlier [Alc+23] calculated $SU(d)$ $6j$ symbols with at least two fundamental representations [Alc+23].
- Representation theoretic proof of the Regge symmetries due to Boalch [Boa07]
- Asymptotics of quantum $6j$ symbols Taylor and Woodward [TW06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].

- Alcock-Zeilinger consider the case of $6j$ symbols for $SU(N)$ for wherein two opposing edges are labelled by fundamental irreps of $SU(N)$ so that all vertices are multiplicity one (or zero) [Alc18] TC: They also freely use the birdtrack notation which I enjoy.
- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling’s approximation applied to Racah’s single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Artamonov derives an explicit formula for the $d = 3$ case [Art23], and later to arbitrary d [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension d but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all d [LS23].

B.4 Minkowski’s theorem and tetrahedron geometry

- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Bianchi, Dona, and Speziale uses Minkowski’s theorem to parametrize convex polyhedra in $\mathbb{R}^3$ with specified face areas and the associated Kapovich-Millson phase space (i.e., the *space of shapes of polyhedra with fixed areas* and then considers its connection to loop quantum gravity via the usual Guillemin-Sternberg quantization commutes with reduction philosophy which associates invariants of the quantization of the unreduced phase space to the reduced phase space [BDS11] (notice the connection with [HHR16]).
- Regarding an algorithmic construction for the polyhedra appearing in Minkowski’s theorem and their complexity, see this post: [constructive-algorithm-for-minkowskis-theorem](#)
- Charles quantization of polygon spaces, tetrahedron and semiclassical limits [Cha10]

B.5 Quantization, Poisson geometry & quantum groups

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- Kirillov and Reshetikhin [KR90]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- Henriques and Kamnitzer [HK05]
- A reference which contains the formula for the quantum dimension of a representation of $U_q(\mathfrak{sl}(d))$ labelled by λ , thanks Rinat! [Ais97]. See also [AM98] which clarifies the relation between various parameters. Apparently, the x, v, s parameters are apparently natural parameters to consider in the context of invariants of knots and links. The substitution mentioned in [AM98] is the following

$$s = e^{h/2}, \quad v = e^{-dh/2} = s^{-d}, \quad x = e^{-h/2d} = s^{-\frac{1}{d}}, \quad q = e^h = s^2.$$

This is consistent with [Res87] in the sense that

$$[k] := \frac{s^{+k} - s^{-k}}{s^{+1} - s^{-1}} = \frac{q^{+k/2} - q^{-k/2}}{q^{+1/2} - q^{-1/2}}. \quad (245)$$

- Reshetikhin develops the diagrammatics of the category of representations of $U_q(\mathfrak{g})$ motivated by connections to the developing theory of polynomial invariants of knots and links [Res87].
- Moak investigates a q -Stirling’s formula for the logarithm of the q -factorial [Moa84].

B.6 Eigenvalue inequalities

- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection between Thompson’s inequality and hyperbolic geometry quite explicitly.
- Friedland and Gaubert considers submodular functions associated to operator monotones evaluated on different principal submatrices of a single Hermitian matrix [FG13]. While it is not clear to be yet whether or not one can relate partial sums of matrices to principal submatrices, [FG13] references a paper by Bouhtou, Gaubert, and Sagnol [BGS08] which proves the following family of inequalities for all positive semidefinite $A, B, C \geq 0$ (and $p \in [0, 1]$) [BGS08, Lemma 5.3]:

$$\sum_j \lambda_j (A + B)^p + \sum_j \lambda_j (B + C)^p \geq \sum_j \lambda_j (B)^p + \sum_j \lambda_j (A + B + C)^p. \quad (246)$$

- Audenaert, Hiai, and Petz consider a similar form of “strongly subadditive” functions [AHP10]
- For multivariate inequalities based on idempotents of the symmetric group algebra see Huber’s paper [Hub20].
- For spectral inequalities derived using semidefinite programming, see Huber and Wyderka’s paper [HW22].
- Paksoy, Turkmen, and Zhang produce a large family of polynomial inequalities for positive semidefinite $A, B, C \geq 0$ using the notion of a “generalized matrix function” [PTZ14] (see also [HM21; MM65])

B.7 Uncategorized

- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Regge and Williams [RW00] a relatively recent review of discrete approaches to gravity. The essential idea, in contrast to the differential geometry of general relativity, is to model curvature discretely instead of continuously. In particular, one can think of a triangulation of the surface of a sphere; the curvature of the sphere is encoded in the triangulation when a configuration of three (or more) triangles meet at a common vertex with sum of interior angles deviating from π . It is in this sense that Regge and Williams refers to these types of models as “piecewise-flat”. Contained is an overview of various attempts to generalize discrete theories of three-dimensional gravity to four dimensions [RW00, Section III].